

Adapting and Fine-Tuning AI-Based Pretrained Video Generation Models for Complex Flow Physics

Adaption und Feinabstimmung KI-basierter vortrainierter Videogenerierungsmodelle für komplexe Strömungsphysik

Scientific work for obtaining the academic degree

Master of Science (M.Sc.)

at the TUM School of Engineering and Design of the Technical University of Munich

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Submitted on	September 27, 2026 in Garching

Abstract

This thesis investigates the transfer of the pretrained LTX-Video 2B architecture from natural video generation to surrogate modeling of two-dimensional compressible Euler flows. Physical trajectories are represented as four-channel spatiotemporal sequences with an additional physical conditioning parameter. The model is adapted in two stages: the variational autoencoder (VAE) is first extended and fine-tuned for physical-field reconstruction, followed by fine-tuning the Diffusion Transformer (DiT) through conditional flow matching on the adapted latent representation.

Full VAE fine-tuning achieves a test reconstruction VRMSE of 0.0800, while the corresponding VAE–DiT rollout reaches 0.4984. Thus, accurate reconstruction alone does not guarantee a latent representation suitable for trajectory generation. DiT-guided self-distillation improves the rollout VRMSE to 0.4583. The results highlight the importance of considering both reconstruction quality and downstream generative suitability when transferring pretrained video models to physical simulation.

Zusammenfassung

Diese Arbeit untersucht die Übertragung der vortrainierten LTX-Video-2B- Architektur von der Videogenerierung auf die Surrogatmodellierung zweidimensionaler kompressibler Euler-Strömungen. Physikalische Trajektorien werden als vierkanalige räumlich-zeitliche Sequenzen mit einem zusätzlichen physikalischen Parameter dargestellt. Das Modell wird in zwei Stufen angepasst: Zunächst wird der Variational Autoencoder (VAE) für physikalische Felder erweitert und feinabgestimmt, anschließend der Diffusion Transformer (DiT) mittels bedingtem Flow Matching auf der angepassten latenten Repräsentation.

Die vollständige VAE-Feinabstimmung erreicht eine Rekonstruktions-VRMSE von 0.0800, während die entsprechende VAE–DiT-Pipeline eine Rollout-VRMSE von 0.4984 erreicht. Eine genaue Rekonstruktion garantiert somit keinen für die Trajektoriengenerierung geeigneten latenten Raum. DiT-geführte Selbstdestillation verbessert die Rollout-VRMSE auf 0.4583. Die Ergebnisse zeigen, dass bei der Übertragung vortrainierter Videomodelle sowohl die Rekonstruktionsqualität als auch die generative Eignung berücksichtigt werden müssen.

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Acronyms

BF16	Brain Floating Point 16-bit
CFD	Computational Fluid Dynamics
DDPM	Denoising Diffusion Probabilistic Model
DiT	Diffusion Transformer
FM	Flow Matching
FNO	Fourier Neural Operator
FP32	32-bit Floating Point
GELU	Gaussian Error Linear Unit
GPU	Graphics Processing Unit
HPC	High-Performance Computing
KL	Kullback–Leibler
LDM	Latent Diffusion Model
MLP	Multilayer Perceptron
PDE	Partial Differential Equation
PINN	Physics-Informed Neural Network
RGB	Red–Green–Blue
RMSE	Root-Mean-Square Error
SDS	Score Distillation Sampling
SSIM	Structural Similarity Index Measure
VAE	Variational Autoencoder
VRMSE	Variance-Normalized Root-Mean-Square Error
XPU	Accelerator Processing Unit

Nomenclature

Physical Quantities

ρ	Density
u	Velocity component in the horizontal direction
v	Velocity component in the vertical direction
m_x	Momentum component in the horizontal direction, $m_x = \rho u$
m_y	Momentum component in the vertical direction, $m_y = \rho v$
p	Pressure
E	Total energy density
γ	Ratio of specific heats
$\mathbf{U}$	Conserved state vector of the Euler equations
$\mathbf{F}(\mathbf{U})$	Flux vector in the horizontal direction
$\mathbf{G}(\mathbf{U})$	Flux vector in the vertical direction

Model and Data Notation

$\mathbf{X}_t$	Physical-field state at time step t
$\mathbf{X}_{0:T}$	Complete physical-field trajectory
$\tilde{\mathbf{X}}$	Normalized physical-field trajectory
$\hat{\mathbf{X}}$	Predicted or reconstructed physical-field state
$\mathbf{z}$	Latent representation
$\mathbf{z}_0$	Latent representation of the initial physical state
$\hat{\mathbf{z}}$	Predicted latent representation
$\mathcal{E}_\phi$	VAE encoder parameterized by ϕ
$\mathcal{D}_\psi$	VAE decoder parameterized by ψ
p_θ	Learned conditional generative distribution parameterized by θ
H	Spatial height of a physical field
W	Spatial width of a physical field

T Final physical time-step index

L Final latent time-step index

Optimization and Statistical Quantities

μ_c Global mean of physical channel c

σ_c Global standard deviation of physical channel c

η Learning rate

λ Loss weighting or regularization coefficient

$\mathcal{L}_{\text{rec}}$ VAE reconstruction objective

$\mathcal{L}_{\text{RMSE}}$ Root-mean-square reconstruction loss

$\mathcal{L}_{\text{H1}}$ Gradient-based H^1 seminorm loss

$\mathcal{L}_{\text{SSIM}}$ Structural-similarity reconstruction loss

$\mathcal{L}_{\text{KL}}$ Kullback–Leibler regularization loss

$\mathcal{L}_{\text{SDS}}$ DiT-guided distillation loss

$\mathcal{L}_{\text{FM}}$ Flow-matching objective

Chapter 1

Introduction

1.1 Motivation

Computational fluid dynamics (CFD) is an essential tool for the numerical analysis of fluid flows in a wide range of engineering applications, including aerospace, automotive engineering, and energy systems. By numerically solving the governing equations of fluid motion, CFD simulations provide detailed spatial and temporal information about physical quantities such as density, momentum, pressure, and velocity. For compressible flows in particular, the accurate simulation of complex phenomena such as shock formation and propagation may require high spatial and temporal resolution, resulting in substantial computational cost [22].

This computational demand can become a major limitation when simulations have to be evaluated repeatedly, for example in design-space exploration, optimization, uncertainty quantification, or real-time applications. Machine-learning-based surrogate models provide a potential alternative by learning an approximation of the underlying physical dynamics from simulation data. Once trained, such models can generate predictions considerably faster than repeatedly solving the original numerical problem. A variety of learning-based approaches have therefore been investigated for the surrogate modeling of partial differential equations (PDEs) and fluid dynamics, including neural operators, graph-based simulators, and physics-informed methods [7, 10, 17, 19].

Most existing approaches, however, employ architectures designed specifically for scientific or physical data and are trained primarily on simulation datasets. More recently, pretrained video generative models have also begun to be explored as surrogate models for physical simulation, demonstrating the potential of transferring representations learned from natural videos to spatiotemporal physical dynamics [16]. This motivates the broader question considered in this thesis: to what extent can the representations and spatiotemporal priors of a pretrained generative video model be transferred from natural visual data to compressible physical simulation?

1.2 From Video Generation to Physical Simulation

A time-dependent CFD simulation and a digital video share a useful structural similarity. Both can be represented as sequences of spatially structured two-dimensional states evolving over time. In natural video, the channels of each state describe visual information such as red, green, and blue intensities. In the compressible-flow setting considered in this thesis, they instead represent physical quantities: density, horizontal momentum, vertical momentum, and pressure. A CFD trajectory can therefore be viewed, at the level of data organization, as a multi-channel spatiotemporal sequence analogous to a video. The precise physical-state representation and surrogate modeling problem are defined in chapter 3.

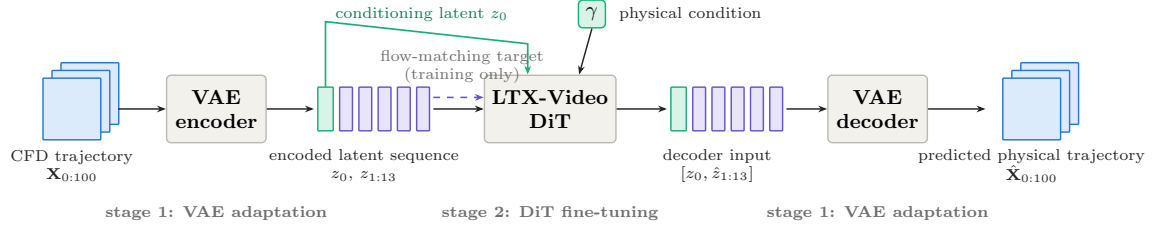


Figure 1.1: Overall concept of the proposed latent generative surrogate framework. A pretrained video-generation architecture is adapted from natural RGB video to multi-channel physical-field trajectories. The VAE provides the compressed spatiotemporal representation, while the DiT models the conditional temporal evolution in latent space.

This structural correspondence motivates the use of architectures originally developed for video generation. Modern latent video models separate high-dimensional data representation from generative modeling: an autoencoding model compresses the input sequence into a latent representation, while a generative model learns its spatiotemporal evolution. The relevant background on latent generative models, Diffusion Transformers, and video generation is reviewed in chapter 2.

This thesis investigates whether such pretrained representations can be repurposed for scientific simulation. Specifically, it builds upon LTX-Video, a pretrained latent video generation architecture [1], and adapts it to compressible Euler flow data. The original three-channel RGB interface is extended to the four physical fields, while scalar physical information, including the ratio of specific heats γ , is introduced as an additional conditioning signal.

The resulting high-level workflow is summarized in fig. 1.1. The adapted VAE provides a compressed representation of the physical trajectory, while the DiT models the conditional temporal evolution in latent space. The generated latent sequence is subsequently decoded back into physical-field trajectories. The detailed architecture and training objectives are introduced later in chapter 4.

Transferring such a model from natural video to CFD is nevertheless non-trivial. Physical fields differ substantially from RGB observations in their dimensionality, statistical properties, semantics, and underlying dynamics. The transfer problem therefore involves two related stages: the VAE must be adapted to represent the physical fields accurately, and the DiT must be adapted to model the temporal evolution of the resulting latent representation.

These stages are not independent. Fine-tuning the VAE encoder for the physical domain changes the representation on which the downstream generative model operates. Consequently, improved VAE reconstruction does not necessarily imply improved trajectory generation. In this thesis, the change in the latent representation induced by domain adaptation is referred to as *representation shift*. More generally, such changes in feature spaces and data distributions are a central consideration in transfer learning [14]. Understanding how VAE adaptation affects downstream DiT performance is therefore a central part of the present study.

1.3 Research Questions

Based on these considerations, this thesis investigates the following research questions:

1. **How can a pretrained video VAE be adapted from natural RGB videos to multi-channel physical simulation data?**

This includes the architectural modifications required for four-channel CFD fields as well as the effects of adaptation strategy, reconstruction objective, learning rate, and spatial resolution.

2. To what extent can a pretrained video Diffusion Transformer be adapted to model the temporal evolution of CFD trajectories?

In particular, the thesis investigates the feasibility of adapting spatiotemporal representations acquired through large-scale video pretraining to the dynamics of compressible flow and evaluates the resulting trajectories after decoding them back into the normalized physical-field representation.

3. How does VAE fine-tuning affect the compatibility between the adapted latent representation and downstream DiT adaptation?

The thesis studies the relationship between VAE reconstruction quality and downstream DiT roll-out performance and investigates regularization and distillation strategies intended to preserve compatibility between the two stages during VAE adaptation.

1.4 Contributions

The main contributions of this thesis are summarized as follows:

- **Adaptation of a pretrained latent video model to compressible CFD data.**

A pretrained video-generation architecture is extended from three-channel natural RGB videos to four-channel physical fields and conditioned on scalar physical parameters. This establishes a two-stage framework for investigating the transfer of a pretrained video generative model to CFD surrogate modeling.

- **Systematic investigation of VAE adaptation.**

The VAE stage is evaluated through controlled ablation experiments covering adaptation strategy, reconstruction and gradient-based objectives, learning-rate sensitivity, and spatial resolution.

- **Analysis of VAE–DiT compatibility.**

The thesis investigates whether improvements in VAE reconstruction quality translate into improved downstream DiT rollout performance. Particular attention is given to the dependency between the representation learned by the adapted VAE and downstream latent dynamics modeling. KL regularization and DiT-guided distillation are further evaluated as strategies intended to preserve compatibility between the adapted VAE representation and the downstream DiT.

These contributions aim not only to evaluate the predictive performance of the resulting surrogate model, but also to provide insight into the broader challenge of transferring pretrained generative models from natural visual data to scientific spatiotemporal domains.

1.5 Thesis Structure

The remainder of this thesis is organized as follows. Chapter 2 reviews existing approaches to machine-learning-based PDE surrogate modeling, diffusion and video generative models, and transfer learning of pretrained representations. Chapter 3 introduces the compressible Euler flow problem and the dataset used in this work. Chapter 4 presents the adaptation framework, including VAE modification, physical conditioning, DiT fine-tuning, and strategies for maintaining compatibility between the VAE and DiT stages. Chapter 5 describes the experimental setup, training configurations, evaluation metrics, and

controlled ablation protocol. Chapter 6 presents and discusses the experimental results for VAE adaptation, DiT-based trajectory prediction, and VAE–DiT compatibility. Finally, Chapter 7 summarizes the main findings, discusses the limitations of the current approach, and outlines directions for future work.

“

Chapter 2

Literature Review

The approach investigated in this thesis lies at the intersection of learning-based physical modeling, generative video modeling, and transfer learning. Rather than developing a task-specific surrogate architecture from scratch, this work investigates whether the spatiotemporal representations of a pretrained video generative model can be adapted to model the evolution of physical fields.

This chapter reviews the literature relevant to this perspective. First, learning-based approaches for modeling partial differential equations (PDEs) and computational fluid dynamics (CFD) are discussed, including neural operators, graph-based learned simulators, and physics-informed methods. These approaches provide the broader context for using neural networks to approximate physical dynamics, rather than serving as direct performance baselines for the present study. The review then introduces diffusion-based generative modeling, with particular attention to latent diffusion, Diffusion Transformers, and flow matching. Subsequently, video generative models and their ability to represent spatiotemporal data are reviewed. Finally, transfer learning and knowledge distillation are discussed in the context of adapting pretrained representations while maintaining compatibility between the VAE and downstream generative model. Representation shift is introduced as a potential consequence of adapting the VAE encoder to the physical domain.

2.1 Learning-Based Approaches for PDE and CFD Modeling

The numerical solution of partial differential equations forms the basis of computational modeling in many areas of science and engineering. In computational fluid dynamics, conventional numerical methods discretize the governing equations in space and time and compute the evolution of the physical state according to the underlying conservation laws [22]. Such numerical methods provide the reference solutions used throughout this thesis. However, high-fidelity simulations can be computationally expensive, particularly when many trajectories must be evaluated across different initial conditions, boundary conditions, geometries, or physical parameters.

This computational cost has motivated extensive research into learning-based approaches for physical simulation. Instead of repeatedly solving the governing equations numerically, these methods learn mappings or temporal evolution rules from simulation data. Depending on the formulation, the learned model may approximate a solution operator, predict the temporal evolution of a discretized physical state, or incorporate the governing equations directly into the optimization objective [7, 10, 17, 19].

Several major methodological directions have emerged. Neural operators learn mappings between function spaces and are therefore naturally suited to parametric PDE problems. Graph-based learned simulators represent spatial discretizations explicitly and learn local interactions that determine the evolution of the physical system. Physics-informed neural networks instead introduce physical constraints through the governing differential equations during optimization. The following subsections briefly review these

approaches to establish the broader context in which the video-model-based approach of this thesis is situated.

2.1.1 Neural Operators

Neural operators aim to learn mappings between function spaces rather than mappings between fixed-dimensional vectors. For a parametric PDE, this allows a model to approximate the solution operator that maps an input function, such as an initial condition, coefficient field, or forcing term, to the corresponding solution field. Once trained, the learned operator can be evaluated for previously unseen instances of the underlying PDE problem.

A representative approach is the Fourier Neural Operator (FNO) proposed by Li et al. [7]. Instead of parameterizing the operator exclusively through local transformations in physical space, FNO performs learned transformations in Fourier space, allowing global interactions to be represented efficiently using a restricted set of Fourier modes. The original work demonstrated the approach on several PDE problems, including Burgers' equation, Darcy flow, and the Navier–Stokes equations. Neural operators are particularly attractive for spatially distributed physical systems because the learned mapping is formulated at the level of functions rather than as a mapping between individual fixed-dimensional state vectors.

DeepONet provides another influential formulation of operator learning. Lu et al. [10] introduced an architecture consisting of a *branch network*, which encodes observations of an input function, and a *trunk network*, which represents the coordinates at which the output function is evaluated. The outputs of the two networks are combined to approximate a nonlinear operator. The original study demonstrated the framework on a range of explicit operators as well as operators associated with deterministic and stochastic differential equations.

Neural operators demonstrate that complex PDE solution mappings can be learned directly from numerical data. At the same time, architectures such as FNO and DeepONet are designed specifically for operator-learning problems and typically acquire their representations from data belonging to the target scientific domain. The present work investigates a different question: whether representations originally learned for natural video generation can be transferred to the modeling of physical-field dynamics.

2.1.2 Graph-Based Learned Simulators

A second class of methods represents physical systems using graphs. This formulation is particularly useful when the simulation domain is naturally described by an irregular mesh, since mesh nodes can be represented as graph vertices and their spatial connectivity as graph edges. Graph neural networks can then propagate information through message passing and learn how the physical state evolves over time.

MeshGraphNets, introduced by Pfaff et al. [17], are a representative example of this approach. The framework learns dynamics directly on mesh-based representations and was demonstrated on multiple physical systems, including aerodynamics, structural mechanics, and cloth simulation. The learned simulator predicts changes in the physical state and repeatedly applies these predictions to obtain an autoregressive rollout of the system.

Such learned simulators provide a natural mechanism for predicting time-dependent physical processes and can operate directly on spatial discretizations used by numerical solvers. Their formulation nevertheless differs substantially from the approach investigated in this thesis. Whereas a graph-based simulator explicitly represents spatial connectivity and predicts state evolution through learned message passing, the present work maps physical fields into the latent representation of a pretrained video model. Spatial and temporal dependencies are therefore modeled through a representation originally learned for video generation rather than through an explicitly constructed mesh graph.

2.1.3 Physics-Informed Neural Networks

Physics-informed neural networks (PINNs) follow a different paradigm. Rather than learning solely from pairs of simulated inputs and outputs, PINNs explicitly incorporate the governing physical equations into the optimization objective. Raissi et al. [19] introduced a general framework in which automatic differentiation is used to evaluate derivatives of the neural network prediction, allowing the residual of the governing PDE to be included as a training loss. A simplified PINN objective can be expressed as

$$\mathcal{L}_{\text{PINN}} = \lambda_{\text{data}} \mathcal{L}_{\text{data}} + \lambda_{\text{PDE}} \mathcal{L}_{\text{PDE}} + \lambda_{\text{BC}} \mathcal{L}_{\text{BC}}, \quad (2.1)$$

where the individual terms penalize disagreement with available observations, violation of the governing equations, and violation of boundary or initial conditions, respectively.

This explicit incorporation of physical knowledge distinguishes PINNs from purely data-driven approaches to physical modeling and can be useful when physical constraints are available independently of simulation observations. The approach pursued in this thesis is fundamentally different. The Euler equations are not imposed through a PDE-residual loss during model training. Instead, trajectories produced by numerical simulation are treated as spatiotemporal training data, and the model learns their evolution from these examples. Physical information enters through the physical-field representation, physical-parameter conditioning, and selected structure-sensitive reconstruction objectives rather than through direct enforcement of the complete governing PDE.

2.1.4 Positioning of This Work

The methods reviewed above illustrate several established ways of applying machine learning to PDEs and physical simulation. Neural operators learn solution mappings between function spaces, graph-based simulators learn temporal dynamics on explicit spatial discretizations, and PINNs incorporate governing equations directly into the optimization process. These approaches provide important evidence that neural models can represent complex physical dynamics, but they generally rely on architectures or learning objectives developed specifically for scientific problems.

This thesis explores a complementary direction based on transfer learning from generative video models. Rather than introducing a new task-specific architecture for the Euler equations, it starts from a latent video model pretrained on natural visual data and investigates how its representations can be adapted to multi-channel physical fields. A simulated flow trajectory is interpreted as a spatiotemporal signal analogous, at the level of data organization, to a video: each time step contains spatially structured fields, while successive time steps describe their temporal evolution. This analogy motivates the reuse of architectures designed to compress and generate temporally coherent video sequences.

The objective of this thesis is therefore not to establish a direct performance ranking against specialized PDE-learning architectures such as FNO, DeepONet, or MeshGraphNets. Instead, these methods define the broader scientific context for learning physical dynamics. The central question considered here is whether spatiotemporal representations acquired through large-scale video pretraining remain useful after a substantial domain shift from natural RGB imagery to compressible flow fields, and what adaptations are required to make the different components of the pretrained latent generative model operate consistently in this new domain.

Addressing this question requires considering not only physical modeling but also the structure of modern latent generative models. In particular, the investigated model separates representation learning and generation into a variational autoencoder and a latent generative model. Consequently, successful transfer requires not only adapting the representation of the physical fields, but also considering how changes to this representation affect the downstream latent generative model. The following sections therefore review diffusion models, latent generative modeling, Diffusion Transformers, and flow matching before turning to video generation and representation transfer.

2.2 Diffusion Models and Flow Matching

Diffusion-based generative modeling provides a powerful framework for learning complex high-dimensional data distributions [3]. Rather than directly constructing samples in a single prediction step, diffusion models learn a gradual transformation between a data distribution and a simple reference distribution, typically a Gaussian distribution. Their success in image and video generation has made them an important foundation for modern generative models and motivates their investigation for other forms of structured spatiotemporal data.

The model investigated in this thesis combines several developments originating from this line of research. High-dimensional observations are first compressed into a latent representation, the generative backbone operates on these latents using a transformer architecture, and the generative objective follows a flow-matching formulation rather than the classical noise-prediction objective of denoising diffusion probabilistic models. The following subsections introduce these concepts in this order and establish the theoretical background required for the pretrained video model considered later in this chapter.

2.2.1 Denoising Diffusion Models

Denoising diffusion probabilistic models (DDPMs), popularized by Ho et al. [3], define a generative process through a sequence of gradual noising and denoising operations. During the forward process, Gaussian noise is progressively added to a data sample $\mathbf{x}_0$ according to a predefined noise schedule. A sample at diffusion step t can be written as

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (2.2)$$

where $\bar{\alpha}_t$ determines the amount of the original signal retained at step t . As the noise level increases, the distribution of $\mathbf{x}_t$ approaches a Gaussian reference distribution.

Generation requires reversing this corruption process. A neural network is trained to estimate information that allows a noisy sample to be progressively transformed toward the data distribution. In the commonly used noise-prediction parameterization, the network ϵ_θ predicts the Gaussian noise contained in $\mathbf{x}_t$. A simplified training objective is

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{\mathbf{x}_0, t, \epsilon} [\|\epsilon - \epsilon_\theta(\mathbf{x}_t, t)\|_2^2]. \quad (2.3)$$

At inference time, sampling begins from Gaussian noise and repeatedly applies the learned reverse process to obtain a sample from the learned data distribution. This iterative construction allows diffusion models to represent complex and multimodal distributions. Applying the generative process directly to high-dimensional image or video data, however, can require substantial computation. This motivates performing generative modeling in a compressed latent space.

2.2.2 Latent Diffusion Models

Latent diffusion models (LDMs), introduced by Rombach et al. [20], reduce the computational cost of generative modeling by separating representation learning from the generative process. An autoencoder first maps a high-dimensional observation $\mathbf{x}$ into a lower-dimensional latent representation,

$$\mathbf{z} = E(\mathbf{x}), \quad \hat{\mathbf{x}} = D(\mathbf{z}), \quad (2.4)$$

where E and D denote the encoder and decoder, respectively. The generative process is then learned in the latent space rather than directly in the original observation space.

This decomposition assigns two distinct roles to the model components. The autoencoder defines a compressed representation that retains information needed for reconstruction, whereas the latent generative model learns to operate on representations produced by the encoder. Conceptually, the generation pipeline can therefore be written as

$$\mathbf{x} \xrightarrow{E} \mathbf{z} \xrightarrow{\text{latent generative model}} \hat{\mathbf{z}} \xrightarrow{D} \hat{\mathbf{x}}. \quad (2.5)$$

Reducing the spatial and temporal dimensionality before applying the generative model can substantially decrease the computational cost of modeling high-dimensional observations. At the same time, this decomposition introduces a dependency between the representation model and the downstream generative model: the latter learns to operate on the particular representation produced by the encoder. This dependency becomes especially relevant in the transfer-learning setting considered in this thesis. When the representation model is adapted to a substantially different target domain, the statistical and structural properties of the representations presented to the downstream model may also change. Accurate reconstruction by the adapted autoencoder therefore does not, by itself, establish that the resulting representation will be equally suitable for downstream latent generation. This possibility motivates the functional compatibility analysis developed later in this thesis.

The investigated video architecture uses a pretrained variational autoencoder (VAE) to compress spatiotemporal data and a separate latent generative model to operate on the resulting representation. Transferring this architecture from natural videos to physical fields therefore requires adapting both components while accounting for their architectural dependency. This issue is discussed further in section 2.4.

2.2.3 Diffusion Transformers

The neural network used to predict the dynamics of the intermediate generative state is another important architectural choice. Early diffusion models commonly employed convolutional U-Net architectures for this purpose. Peebles and Xie [15] demonstrated that the U-Net backbone can instead be replaced by a transformer operating on latent patches, introducing the Diffusion Transformer (DiT).

In a DiT, the latent representation is partitioned into patches and projected into a sequence of tokens. These tokens are processed by transformer blocks, while information about the diffusion or flow timestep and other conditioning variables is incorporated into the network. Through self-attention, tokens can exchange information across the sequence, allowing the model to represent long-range dependencies within the latent representation.

The original DiT study showed that increasing transformer computation through model depth, width, and token count was associated with improved generative performance [15]. Transformer-based generative backbones have subsequently become important components of large-scale image and video generation systems.

For the setting considered in this thesis, the significance of the DiT architecture is that it provides the generative component operating on the compressed spatiotemporal representation. A video trajectory contains dependencies across both spatial locations and temporal steps, and a transformer-based latent model provides a flexible mechanism for representing such interactions. The pretrained video model considered later in this chapter uses such a transformer-based generative component. This thesis investigates whether this component can be adapted from latent representations of natural videos to latent representations of physical-field trajectories.

2.2.4 Flow Matching

The term diffusion model is commonly associated with the stochastic noising and reverse-denoising formulation described above. Modern generative models can, however, employ related formulations that learn continuous transformations between probability distributions. Flow Matching, introduced by Lipman et al. [8], formulates generative modeling in terms of a time-dependent vector field associated with a prescribed probability path between a data distribution and a simple reference distribution. Consider a latent data sample $\mathbf{z}_0$ and an independent Gaussian sample ϵ . Using the convention adopted in this thesis, a linear path from data at $t = 0$ to noise at $t = 1$ is defined as

$$\mathbf{z}_t = (1 - t)\mathbf{z}_0 + t\epsilon, \quad t \in [0, 1]. \quad (2.6)$$

For this linear interpolation, the corresponding target velocity is

$$\mathbf{v} = \frac{d\mathbf{z}_t}{dt} = \epsilon - \mathbf{z}_0. \quad (2.7)$$

A neural network $\mathbf{v}_\theta$ is trained to predict this velocity from the intermediate state $\mathbf{z}_t$ and the corresponding time t . A simplified flow-matching objective can be written as

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{\mathbf{z}_0, \epsilon, t} [\|\mathbf{v}_\theta(\mathbf{z}_t, t) - (\epsilon - \mathbf{z}_0)\|_2^2]. \quad (2.8)$$

With this convention, the interpolation used during training is parameterized from data at $t = 0$ to noise at $t = 1$. Generation proceeds in the reverse direction: an initial sample is drawn from the reference distribution and the learned vector field is integrated from the noise endpoint toward the data endpoint. Equivalent formulations may reverse the time convention and therefore the sign of the target velocity; the training target and sampling dynamics must therefore be interpreted consistently.

Compared with the classical DDPM noise-prediction formulation, flow matching directly trains a vector field associated with a chosen probability path [8]. The two formulations differ in their mathematical construction and training target, but both provide mechanisms for learning transformations between a simple reference distribution and a complex data distribution.

Flow matching is the formulation directly used by the latent generative model in this thesis. After the VAE has been adapted to physical fields, CFD trajectories are encoded into latent representations, and the transformer is fine-tuned to predict the corresponding flow-matching velocities. The flow-matching objective is therefore defined in latent space.

Because this objective directly evaluates the transformer’s prediction with respect to a latent training target, it does not by itself characterize the quality of the complete generative pipeline in the original observation space. The final output additionally depends on the representation and decoding behavior of the VAE. Evaluation in the normalized physical-field representation is therefore required to assess the complete surrogate model.

Together, latent representation learning, transformer-based generative modeling, and flow matching provide the principal architectural and mathematical foundations for the pretrained video model used in this work. The following section extends this discussion to video generative models, where these components are combined to represent spatial and temporal structure.

2.3 Video Generative Models

The generative modeling principles introduced in the previous section can be extended from individual images to videos. Compared with image generation, video generation introduces an additional temporal

dimension and therefore requires a model to represent not only spatial structure within individual frames but also dependencies across time [4]. This makes video generative models particularly relevant to the question investigated in this thesis: whether representations learned for natural spatiotemporal data can be transferred to the evolution of physical fields.

This section first introduces the general formulation of video diffusion models. It then discusses LTX-Video, the pretrained latent video model that forms the architectural basis of this work, followed by WinDiNet, a direct predecessor that applies a related transfer-learning approach to wind-flow data. Finally, the relationship between video generation and physical-field modeling is discussed to motivate the transfer-learning perspective adopted in this thesis.

2.3.1 Video Diffusion Models

Video diffusion models extend diffusion-based generative modeling from spatial images to spatiotemporal sequences. Ho et al. [4] demonstrated that diffusion architectures can be extended to video by introducing temporal modeling in addition to the spatial processing used for individual images. Rather than generating frames independently, the model represents dependencies across both space and time, enabling the generation of temporally coherent sequences.

A video can be represented as a tensor

$$\mathbf{X} \in \mathbb{R}^{T \times C \times H \times W}, \quad (2.9)$$

where T denotes the number of frames, C the number of channels, and H and W the spatial dimensions. The additional temporal dimension substantially increases the dimensionality of the generative problem compared with image generation. Modern video generation systems can combine temporal modeling with latent-space compression, such that the computationally expensive generative process operates on a lower-dimensional spatiotemporal representation rather than directly on the original video; LTX-Video is one example of this approach [1].

A time-dependent physical simulation admits a structurally similar organization: each time step contains multiple spatially distributed physical fields, while successive states form a temporal sequence. In the compressible flow setting considered in this thesis, these channels correspond to density, two momentum components, and pressure. Both videos and physical trajectories can therefore be organized as sequences of spatially structured multi-channel states.

This correspondence is structural rather than semantic. RGB channels describe visual appearance, whereas physical channels represent quantities whose evolution is governed by coupled conservation laws. In addition to this semantic difference, the pretrained model operates on three-channel RGB observations, whereas the physical state considered in this thesis consists of four distinct channels. The transfer problem therefore involves both a change in channel dimensionality and a change in the meaning and statistical properties of the represented quantities.

Similarly, temporal variation in natural videos may result from object motion, camera motion, illumination changes, and other visual phenomena, whereas the evolution of a CFD trajectory follows the governing physical dynamics. Consequently, similarity in tensor structure alone does not imply that representations learned from natural videos are directly suitable for physical simulation.

Nevertheless, the shared spatiotemporal organization motivates the transfer hypothesis considered in this work: representations acquired through large-scale video pretraining may contain spatial and temporal structure that can be adapted to physical-field sequences. Whether and to what extent such representations remain useful after this substantial domain change is an empirical question investigated in the subsequent chapters.

2.3.2 LTX-Video

LTX-Video [1] is a transformer-based latent video generation model designed to perform video generation in a strongly compressed spatiotemporal latent space. Its architecture follows the latent generative paradigm described in section 2.2.2: a Video-VAE compresses the input video into a latent representation, while a transformer-based generative model operates on the resulting latent sequence.

At a conceptual level, the architecture can be written as

$$\mathbf{X} \xrightarrow{E_{\text{VAE}}} \mathbf{Z} \xrightarrow{\text{DiT}} \hat{\mathbf{Z}} \xrightarrow{D_{\text{VAE}}} \hat{\mathbf{X}}, \quad (2.10)$$

where $\mathbf{X}$ denotes the input spatiotemporal signal, $\mathbf{Z}$ its compressed latent representation, and $\hat{\mathbf{X}}$ the output obtained after latent generation and decoding. The VAE and transformer therefore fulfill distinct but interdependent roles. The VAE determines how high-dimensional observations are represented in latent space, whereas the transformer models the generative dynamics of the resulting latent spatiotemporal sequences.

LTX-Video employs strong spatiotemporal compression in its Video-VAE. The architecture uses spatial downscaling by a factor of 32 in each spatial dimension and temporal downscaling by a factor of 8, corresponding to $32 \times 32 \times 8$ input pixels per latent token [1]. Consequently, a spatial resolution of 128×128 corresponds to a 4×4 latent spatial grid, while a 256×256 input corresponds to an 8×8 grid. Such compression substantially reduces the number of latent tokens that must be processed by the generative model.

The generative component is transformer-based and operates on this compressed spatiotemporal representation [1]. The model employs a flow-matching-based generative formulation, connecting the architecture to the framework introduced in section 2.2.4. The transformer therefore models a velocity field over intermediate latent states rather than following the classical DDPM noise-prediction formulation.

The separation between representation learning and latent generation is particularly important when transferring the model to a new domain. The transformer does not operate directly on RGB values or physical quantities; it operates on representations produced by the VAE encoder. Consequently, the VAE and transformer cannot be treated as entirely independent components. Adapting the encoder changes the representation presented to the transformer, while the decoder determines how generated latent states are mapped back into the observation domain.

2.3.3 WinDiNet

A direct predecessor of the present work is WinDiNet [16], which repurposes a pretrained video generative model as a surrogate for time-dependent wind-flow simulation. Starting from LTX-Video, the model is adapted using two-dimensional incompressible CFD trajectories, providing a direct example of transferring a pretrained latent video architecture from natural visual data to physical flow data.

The motivation underlying WinDiNet is closely related to the transfer-learning perspective of this thesis. Physical trajectories and natural videos differ substantially in their semantics and governing mechanisms, but both consist of spatially structured states evolving over time. WinDiNet therefore provides a direct precedent for exploiting this structural correspondence by adapting a pretrained video model to physical flow simulation.

The present thesis builds on this direction and investigates transfer to compressible Euler-flow trajectories. In contrast to the two predicted velocity components in the preceding wind-flow setting, the target physical state considered here contains four coupled variables: density, horizontal momentum, vertical momentum, and pressure. The considered trajectories may also contain strong gradients and discontinuous structures characteristic of compressible flow, including shock waves.

These differences introduce additional adaptation requirements. In particular, the original video representation is based on three-channel RGB observations, whereas the physical state considered here

contains four channels with distinct physical semantics. The pretrained VAE must therefore accommodate both a different channel dimensionality and substantially different field statistics. At the same time, the latent generative component must model the coupled temporal evolution of multiple physical quantities. Moreover, because the VAE and latent generative model are adapted as separate stages, the effect of VAE adaptation on downstream generative performance becomes an important consideration. The present work therefore extends the direction established by WinDiNet toward a substantially different compressible-flow domain while examining the interaction between representation adaptation and latent trajectory modeling.

2.3.4 Transfer from Video Generation to Physical-Field Modeling

The use of pretrained video models for physical-field modeling differs conceptually from the purpose-built learning approaches discussed in section 2.1. Neural operators, graph-based simulators, and physics-informed methods are designed specifically around properties of scientific problems. A pretrained video model instead begins with representations acquired from a different source domain and relies on transfer learning to adapt them to physical data [14].

The motivation for this approach is that large-scale video pretraining may provide reusable representations of spatial organization and temporal variation. Rather than learning every aspect of spatiotemporal representation from the target simulation dataset alone, transfer learning begins from a model that has already acquired substantial representational capacity from natural video. Whether this capacity can be effectively reused for physical fields, however, is an empirical question rather than an assumption of this thesis.

The domain shift involved is substantial. Natural-video representations are not explicitly designed to encode conservation laws, shock propagation, or coupled thermodynamic variables. The statistical distributions and semantics of physical channels also differ fundamentally from those of RGB image channels. For the present problem, the change from three RGB channels to four physical fields additionally requires an architectural adaptation of the observation interface. Consequently, transferring the pretrained video model to CFD requires adaptation of both its physical-field representation and its latent generative dynamics.

A further challenge arises from the multi-component structure of latent video models. The VAE defines the representation on which the transformer operates. Adapting the VAE to a new domain can therefore affect not only reconstruction quality but also the inputs presented to the downstream latent generative model. Evaluation of the representation stage in isolation is therefore insufficient to characterize the complete surrogate.

This observation motivates treating VAE–DiT compatibility as part of the transfer problem. Changes in the representation induced by VAE fine-tuning provide one possible mechanism through which downstream performance may be affected. The broader conceptual context for this interaction is discussed in the following section.

2.4 Transfer Learning and Representation Compatibility

Adapting a pretrained video model to physical simulation constitutes a transfer-learning problem. Rather than learning all model parameters exclusively from the target CFD dataset, the model begins from representations and generative behavior acquired during large-scale pretraining on a different source domain. In general, transfer learning concerns the reuse of knowledge acquired in a source task or domain to support learning in a different target task or domain [14].

The setting considered in this thesis introduces an additional challenge. Modern latent generative models consist of multiple interacting components. In particular, the VAE defines the latent representation,

while the downstream generative model is trained to operate on that representation. Adapting one component can therefore modify the interface presented to another pretrained component. Successful transfer consequently depends not only on adapting individual components to the target domain but also on maintaining sufficient compatibility between them.

This section first reviews transfer learning and representation adaptation in the context of substantial domain change. It then discusses how adaptation of one component can affect downstream compatibility in modular pretrained architectures. Knowledge distillation is subsequently introduced as a general mechanism for transferring behavior from a teacher model. Together, these concepts provide the background for the VAE–DiT compatibility problem investigated in later chapters.

2.4.1 Transfer Learning and Representation Adaptation

Transfer learning aims to reuse knowledge acquired from a source task or domain when learning a different target task or domain [14]. A common strategy is to initialize a model with pretrained parameters and subsequently fine-tune some or all of its layers on target-domain data. The general objective is to exploit knowledge acquired from the source setting rather than learning the target problem entirely from scratch.

The effectiveness of transfer depends on the relationship between the source and target domains. Differences in feature spaces, marginal distributions, tasks, and domains form part of the standard transfer-learning formulation [14]. Under a substantial domain change, the pretrained representation may therefore require significant adaptation before it becomes suitable for the target data.

The domain shift investigated in this thesis is substantial. The source domain consists of natural three-channel RGB videos, whereas the target domain consists of four-channel numerical fields representing compressible fluid dynamics. Although both can be organized as spatiotemporal tensors, their channel dimensionality, channel semantics, value distributions, spatial structures, and temporal dynamics differ considerably. The pretrained representation must therefore be adapted before it can provide an effective encoding of the physical data.

For a single-component model, adaptation can often be evaluated primarily through performance on the target task. In a multi-component latent model, this criterion is not necessarily sufficient. Fine-tuning the VAE may improve the reconstruction of target-domain observations while simultaneously modifying the representation supplied to the downstream latent generative model. Representation quality must therefore be considered not only with respect to the objective of the adapted component but also with respect to its role within the complete pretrained system.

2.4.2 Representation Adaptation and Downstream Compatibility

Fine-tuning can modify parameters and internal representations of a neural network. A related phenomenon has been studied extensively in continual learning, where adaptation to new tasks can interfere with previously acquired behavior, leading in some settings to catastrophic interference or forgetting [6, 12]. The concern in the present setting, however, differs from conventional catastrophic forgetting. Preserving the VAE’s ability to reconstruct natural videos is not the objective. Instead, the relevant question is whether adaptation of the VAE changes its representation in ways that affect the downstream latent generative model.

Consider a pretrained encoder E_{pre} and an encoder E_{ft} obtained after adaptation to the target physical domain. For the same input $\mathbf{x}$, they produce representations

$$\mathbf{z}_{\text{pre}} = E_{\text{pre}}(\mathbf{x}), \quad \mathbf{z}_{\text{ft}} = E_{\text{ft}}(\mathbf{x}). \quad (2.11)$$

There is no constraint in ordinary reconstruction-oriented fine-tuning that requires these two representations to retain identical statistical or structural properties. The adapted encoder may learn features that are more suitable for reconstructing the target-domain data while differing from the representation associated with the pretrained generative model. In this thesis, such changes are referred to as *representation shift* induced by domain adaptation.

This creates a potential mismatch between the local objective used to optimize the VAE and the requirements of the latent generative model. In the setting of this thesis, the DiT is subsequently fine-tuned on the adapted CFD latents. The relevant question is therefore not whether a frozen pretrained DiT can consume the adapted representation without modification, but whether the adapted representation remains suitable for effective downstream DiT adaptation.

This distinction is important for modular pretrained architectures. A component can improve according to its local objective without necessarily producing a corresponding improvement in the complete system. It is therefore useful to distinguish between *component-level quality*, such as VAE reconstruction accuracy, and *downstream compatibility*, which in this thesis refers to how effectively the representation produced by the adapted component supports subsequent latent dynamics modeling.

Representation shift provides one possible explanation for changes in downstream compatibility. However, a change in representation is not inherently harmful: different representations may encode the information required by the downstream task in different ways. Conversely, similarity in simple latent statistics does not necessarily imply functional compatibility with a particular generative model. For this reason, the present thesis does not attempt to define compatibility through a separate quantitative measure of representation shift. Instead, compatibility is assessed functionally by examining how different VAE adaptations affect subsequent DiT training and the accuracy of the final decoded physical-field trajectories.

2.4.3 Knowledge Distillation

Knowledge distillation was introduced as a method for transferring knowledge from a teacher model to a student model by training the student using information provided by the teacher in addition to conventional supervision [2]. A generic auxiliary-distillation formulation can be written as

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda_{\text{distill}} \mathcal{L}_{\text{distill}}, \quad (2.12)$$

where $\mathcal{L}_{\text{task}}$ optimizes performance on the target task and $\mathcal{L}_{\text{distill}}$ represents an auxiliary teacher-derived objective.

The original motivation for knowledge distillation concerns transferring behavior from a teacher to a student model [2]. The broader teacher–student principle motivates its use here as a conceptual basis for constructing an auxiliary signal from a pretrained downstream component. This use should be distinguished from the conventional model-compression setting of knowledge distillation.

For the flow-based generative model considered in this thesis, the predicted velocity field provides one possible functional signal. Given intermediate latent states, a pretrained DiT produces velocity predictions according to its learned generative dynamics. This motivates using those predictions as an auxiliary teacher signal during VAE adaptation, thereby introducing information from the downstream generative component into the otherwise reconstruction-oriented VAE optimization.

This principle motivates the DiT-guided distillation strategy investigated in this thesis. During VAE adaptation, a frozen pretrained DiT provides an auxiliary signal in addition to the target-domain reconstruction objective. The purpose of this experiment is to investigate whether incorporating downstream-model information during VAE adaptation can improve the interaction between the adapted VAE and the subsequently fine-tuned DiT. This is a methodological hypothesis of the present work rather than a consequence assumed from conventional knowledge distillation. The specific formulation and implementation

are introduced in chapter 4, while its effect on VAE reconstruction and downstream physical-field generation is evaluated in chapter 6.

2.4.4 Research Gap and Relation to This Thesis

The literature reviewed in this chapter establishes several complementary research directions. Learning-based physical models demonstrate that neural networks can represent complex PDE solution mappings and dynamical systems. Diffusion and flow-based generative models provide methods for learning high-dimensional data distributions, while latent video models extend these ideas to compressed spatiotemporal representations. Transfer learning further provides a mechanism for reusing knowledge acquired through pretraining in new domains.

The transfer of a multi-component latent video model to physical-field dynamics combines these directions in a setting where representation adaptation and generative modeling are closely coupled. Adapting the VAE is necessary because the target physical data differ substantially from natural RGB video, including the change from three visual channels to four coupled physical fields. At the same time, the VAE defines the latent representation on which the generative component operates. Improving the VAE for target-domain reconstruction can therefore change the conditions under which the downstream DiT must subsequently learn physical-field dynamics.

This creates a research problem that cannot be reduced to autoencoder reconstruction quality alone. An effective adapted representation must encode the target physical fields with sufficient fidelity while also supporting effective latent dynamics modeling by the downstream transformer. These two requirements need not be perfectly aligned: optimizing the VAE according to its local reconstruction objective does not necessarily optimize the performance of the complete VAE–DiT surrogate.

The present thesis addresses this problem by evaluating adaptation at both levels. VAE reconstruction performance is first used to assess the quality of physical-field representation. The adapted VAE is then combined with DiT fine-tuning, and the resulting trajectories are evaluated after decoding into the normalized physical-field representation. The relationship between these two levels provides a functional assessment of VAE–DiT compatibility without requiring a separate metric for representation shift.

Finally, the thesis investigates whether compatibility can be influenced during VAE adaptation. In addition to reconstruction-oriented objectives, KL regularization and DiT-guided distillation are considered as compatibility-oriented regularization strategies. Their usefulness is evaluated empirically through both VAE reconstruction and downstream DiT performance rather than assumed from changes in the latent representation alone.

Together, these questions define the transition from the literature reviewed in this chapter to the methodology developed in the following chapters. The next chapter first introduces the compressible Euler problem and the physical dataset used in this work, after which the complete VAE–DiT adaptation framework and training objectives are presented. ““

Chapter 3

Problem Formulation and Dataset

This chapter defines the physical problem and dataset used throughout this thesis. First, the governing equations of the considered compressible flow problem are introduced, including the representation of the physical state used for machine-learning-based surrogate modeling. Subsequently, the surrogate modeling objective is formulated as a conditional spatiotemporal generation problem. Finally, the Euler Multi-Quadrants dataset is described, including its physical variables, simulation parameters, preprocessing procedure, and data split used for the experiments.

3.1 Compressible Euler Equations

This thesis considers two-dimensional compressible inviscid flows governed by the Euler equations. The Euler equations describe the conservation of mass, momentum, and energy in a compressible fluid and are able to represent complex phenomena such as shock waves, contact discontinuities, and rarefaction waves that frequently occur in high-speed flows [22].

The conservative form of the two-dimensional Euler equations can be written as

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}(\mathbf{U})}{\partial x} + \frac{\partial \mathbf{G}(\mathbf{U})}{\partial y} = 0, \quad (3.1)$$

where the conserved state vector is defined as

$$\mathbf{U} = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ E \end{bmatrix}. \quad (3.2)$$

Here, ρ denotes density, u and v represent the velocity components in the horizontal and vertical directions, respectively, and E denotes the total energy density. The corresponding flux vectors are

$$\mathbf{F}(\mathbf{U}) = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ u(E + p) \end{bmatrix}, \quad (3.3)$$

and

$$\mathbf{G}(\mathbf{U}) = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ v(E + p) \end{bmatrix}. \quad (3.4)$$

The closure relation between pressure and conserved variables is provided by the ideal-gas equation of state:

$$p = (\gamma - 1) \left(E - \frac{1}{2} \rho (u^2 + v^2) \right), \quad (3.5)$$

where γ is the ratio of specific heats. Different values of γ correspond to different thermodynamic properties of the simulated fluid.

Although the conservative Euler formulation naturally contains density, momentum, and energy, the machine-learning model in this thesis operates on a different physical representation. Specifically, each simulation state is represented by four physical fields:

$$\mathbf{X}_t = \begin{bmatrix} \rho_t \\ (m_x)_t \\ (m_y)_t \\ p_t \end{bmatrix}, \quad (3.6)$$

where

$$m_x = \rho u, \quad m_y = \rho v \quad (3.7)$$

denote the horizontal and vertical momentum components.

Therefore, instead of using the total energy field E , the pressure field p is directly included as one of the model channels. This representation provides four spatially distributed physical quantities:

$$[\rho, m_x, m_y, p].$$

These four fields form the observation space used throughout this thesis.

The scalar parameter γ is not included as an additional spatial channel. Within an individual simulation trajectory, γ remains constant over space and time. Instead, it is treated as an external scalar physical condition provided to the model.

The considered compressible Euler problem is challenging for data-driven modeling because the physical fields are strongly coupled through conservation laws. In addition, the solution may contain sharp discontinuities and rapidly changing structures, including shock waves and contact interfaces. A successful surrogate model must therefore capture both the spatial organization of the physical fields and their temporal evolution.

3.2 Surrogate Modeling Objective

The objective of this thesis is to construct a machine-learning-based surrogate model that predicts the evolution of compressible flow trajectories. Instead of performing an expensive numerical simulation for

every initial condition, the model learns the mapping between the initial physical state and the future trajectory from previously generated simulation data.

A single physical state is represented as

$$\mathbf{X}_t \in \mathbb{R}^{4 \times H \times W}, \quad (3.8)$$

where the four channels correspond to density, horizontal momentum, vertical momentum, and pressure. A complete simulation trajectory is defined as

$$\mathbf{X}_{0:T} = (\mathbf{X}_0, \mathbf{X}_1, \dots, \mathbf{X}_T), \quad (3.9)$$

which can be interpreted as a multi-channel video sequence. While a natural video consists of three RGB channels,

$$\mathbb{R}^{T \times 3 \times H \times W},$$

the physical trajectory considered in this thesis consists of four coupled physical channels,

$$\mathbb{R}^{T \times 4 \times H \times W}.$$

This correspondence motivates the adaptation of pretrained video generative models to physical simulation.

The dataset is formulated as

$$\mathcal{D} = \left\{ (\mathbf{X}_{0:T}^{(i)}, \gamma^{(i)}) \right\}_{i=1}^N, \quad (3.10)$$

where each sample contains a complete physical trajectory together with its corresponding scalar thermodynamic parameter.

The surrogate modeling task is formulated as conditional trajectory generation:

$$\hat{\mathbf{X}}_{1:T} \sim p_{\theta}(\mathbf{X}_{1:T} | \mathbf{X}_0, \gamma). \quad (3.11)$$

The initial state $\mathbf{X}_0$ provides the starting flow configuration, while γ provides additional information about the thermodynamic properties of the simulated system.

Unlike autoregressive one-step prediction methods that recursively predict

$$\mathbf{X}_0 \rightarrow \mathbf{X}_1 \rightarrow \mathbf{X}_2 \rightarrow \dots,$$

the formulation considered in this thesis treats the complete future trajectory as the prediction target. The model therefore learns the spatiotemporal evolution of the entire sequence.

The detailed architecture used to achieve this objective, including VAE adaptation, latent trajectory modeling, and physical conditioning, is introduced in the following methodology chapter.

3.3 Euler Multi-Quadrants Dataset

The experiments in this thesis are conducted on the Euler Multi-Quadrants dataset from The Well benchmark [13]. The dataset contains two-dimensional compressible Euler simulations generated using the Clawpack/AMRClaw framework [11, 13].

Each simulation provides several physical fields, including density, pressure, energy, and momentum components. The original dataset therefore contains the variables

$$[\rho, E, p, m_x, m_y].$$

For the experiments in this thesis, the energy field is omitted, and the model operates on the four-channel representation

$$[\rho, m_x, m_y, p].$$

The Euler Multi-Quadrants simulations are generated from initial conditions constructed by assigning different constant physical states to different spatial quadrants. During the temporal evolution, interactions between these regions generate complex wave patterns, including shock waves, rarefaction waves, and contact discontinuities [13].

Each simulation trajectory is associated with a value of the ratio of specific heats γ . The dataset contains ten different values:

$$\gamma \in \{1.13, 1.22, 1.30, 1.33, 1.365, 1.40, 1.404, 1.453, 1.597, 1.76\}. \quad (3.12)$$

Within one trajectory, γ remains constant and is provided as a scalar conditioning variable for the model.

3.3.1 Spatial Resolution

The original Euler Multi-Quadrants simulations are generated on a uniform Cartesian grid with spatial resolution

$$512 \times 512$$

pixels [13].

For the experiments in this thesis, two preprocessed spatial resolutions are used:

$$256 \times 256$$

and

$$128 \times 128.$$

The 256×256 resolution is used as the main experimental setting, while the 128×128 resolution is used for resolution-related ablation studies.

The two resolutions are prepared before training and are treated as separate dataset versions.

3.3.2 Temporal Resolution

Each trajectory used in the experiments contains 101 recorded physical states:

$$\mathbf{X}_{0:100} = (\mathbf{X}_0, \mathbf{X}_1, \dots, \mathbf{X}_{100}). \quad (3.13)$$

This temporal organization corresponds to the preprocessed trajectory representation used throughout the experiments.

No temporal subsampling or sliding-window construction is applied. Each training sample contains the complete trajectory, allowing the model to learn the global spatiotemporal evolution of the simulated flow. The temporal structure is kept identical for both spatial resolutions.

3.3.3 Normalization

Before being provided to the model, each physical channel is normalized independently. For channel c , the normalized value is computed as

$$\tilde{x}_c = \text{clip}\left(\frac{x_c - \mu_c}{5\sigma_c}, -1, 1\right), \quad (3.14)$$

where μ_c and σ_c are the global mean and standard deviation computed separately for each physical channel.

The scaling factor of 5σ maps the majority of values into the interval approximately corresponding to the input range of the pretrained video model, while clipping limits extreme values.

The statistics are calculated independently for the two spatial resolutions. All reconstruction and trajectory prediction metrics reported in later chapters are evaluated in this normalized physical-field space unless stated otherwise.

3.3.4 Dataset Split

The final dataset used in the experiments consists of

4000

training trajectories,

500

validation trajectories, and

500

test trajectories.

The training and validation sets are randomly selected from the available trajectories using a fixed random seed. The test set is kept separate and is only used for final evaluation.

The split is performed at the simulation level rather than at the individual frame level. Therefore, different temporal states from the same simulation trajectory cannot appear in different partitions.

The same train, validation, and test split is used for both VAE adaptation and DiT fine-tuning experiments, ensuring consistent comparison between different adaptation strategies.

Chapter 4

Methodology

This chapter presents the methodology used to adapt a pretrained latent video generative model to conditional trajectory generation for compressible Euler flows. The framework is based on LTX-Video [1] and consists of two principal components: a variational autoencoder (VAE), which maps high-dimensional physical-field trajectories to a compact spatiotemporal latent representation, and a transformer-based latent generative model, referred to throughout this thesis as the diffusion transformer (DiT), which models the conditional evolution of these trajectories in latent space.

The adaptation is performed in two stages. First, the pretrained VAE is adapted from three-channel natural video to four-channel physical-field data. Different VAE adaptation strategies and reconstruction objectives are investigated. Second, trajectories are encoded using a selected adapted VAE, and the pretrained DiT is fine-tuned on the resulting latent representation using conditional flow matching [8].

The DiT is conditioned on both the initial physical state and the trajectory-level ratio of specific heats γ . The initial state is provided through first-frame latent conditioning, while γ is embedded into conditioning tokens and supplied through the transformer's existing conditioning pathway.

A central methodological consideration is that VAE reconstruction quality and downstream latent generative performance are related but distinct. Adapting the VAE changes the representation on which the DiT subsequently operates, and accurate reconstruction alone does not guarantee accurate decoded trajectory generation. This motivates the functional VAE–DiT compatibility analysis and the compatibility-oriented regularization strategies introduced later in this chapter.

4.1 Overall Framework

4.1.1 From Video Generation to Physical-Field Modeling

The pretrained LTX-Video architecture performs generative modeling in a compressed spatiotemporal latent representation rather than directly in the high-dimensional video space [1]. This decomposition is retained when transferring the model to physical-field trajectories. Given a normalized physical trajectory $\tilde{\mathbf{X}}$, the VAE encoder $\mathcal{E}_\phi$ maps it to a latent representation,

$$\mathbf{z} = \mathcal{E}_\phi(\tilde{\mathbf{X}}), \quad (4.1)$$

while the decoder $\mathcal{D}_\psi$ reconstructs the physical trajectory,

$$\hat{\tilde{\mathbf{X}}} = \mathcal{D}_\psi(\mathbf{z}). \quad (4.2)$$

Here, $\tilde{\mathbf{X}}$ denotes the normalized four-channel physical-field representation introduced in chapter 3.

After VAE adaptation, the DiT models the conditional distribution of future latent states,

$$\hat{\mathbf{z}}_{1:L} \sim p_{\theta}(\mathbf{z}_{1:L} \mid \mathbf{z}_0, \gamma), \quad (4.3)$$

where $\mathbf{z}_0$ represents the latent state associated with the initial physical frame and γ is the ratio of specific heats.

The generated latent sequence is subsequently decoded into the normalized physical-field representation,

$$\hat{\mathbf{X}}_{0:T} = \mathcal{D}_{\psi}(\mathbf{z}_0, \hat{\mathbf{z}}_{1:L}). \quad (4.4)$$

The resulting framework therefore retains the latent-video modeling principle of the pretrained architecture while replacing natural RGB observations with multivariate physical fields. The original semantic conditioning interface is repurposed to provide the physical parameter γ .

4.1.2 Adapted LTX-Video Architecture

The adapted framework contains three principal components: an adapted VAE, a scalar-conditioning module, and a fine-tuned DiT. The underlying pretrained VAE–transformer decomposition follows LTX-Video [1], while the observation interface and conditioning representation are modified for the physical domain.

The VAE maps the four physical variables to the latent representation required by the generative transformer. Depending on the adaptation strategy, this is achieved either by modifying the native VAE input and output layers or by introducing learned channel adapters around the pretrained VAE.

The scalar-conditioning module transforms the trajectory-level parameter γ into a sequence of embedding tokens. These tokens replace the text-conditioning representation used by the pretrained video model and are supplied to the DiT through its existing conditioning pathway.

The DiT receives the first latent frame as a clean spatial condition and generates the remaining latent trajectory jointly. The model is therefore not used as an autoregressive one-step predictor. Instead, the complete future latent sequence is generated through the flow-matching process conditioned on the initial state and γ .

The overall framework can be summarized as

$$\begin{aligned} \tilde{\mathbf{X}}_{0:T} &\xrightarrow{\mathcal{E}_{\phi}} \mathbf{z}_{0:L}, \\ (\mathbf{z}_0, \gamma) &\xrightarrow{\text{conditional DiT}} (\mathbf{z}_0, \hat{\mathbf{z}}_{1:L}), \\ (\mathbf{z}_0, \hat{\mathbf{z}}_{1:L}) &\xrightarrow{\mathcal{D}_{\psi}} \hat{\mathbf{X}}_{0:T}. \end{aligned} \quad (4.5)$$

4.1.3 Two-Stage Adaptation Strategy

The model is adapted in two separate stages.

In the first stage, the VAE is fine-tuned for reconstruction of normalized Euler-flow trajectories,

$$\tilde{\mathbf{X}} \xrightarrow{\mathcal{E}_{\phi}} \mathbf{z} \xrightarrow{\mathcal{D}_{\psi}} \hat{\mathbf{X}}. \quad (4.6)$$

Different subsets of the VAE parameters are optimized depending on the adaptation strategy described in section 4.2.

In the second stage, a selected VAE checkpoint is fixed and used to encode the dataset. The resulting latent trajectories are precomputed and subsequently used to fine-tune the DiT. No gradient from the DiT training objective is propagated through the VAE during this stage.

Consequently, each downstream DiT experiment is associated with a specific VAE checkpoint and therefore with the latent representation defined by that checkpoint. This separation makes it possible to first evaluate VAE reconstruction performance and then investigate how the corresponding representation behaves in downstream conditional trajectory generation.

4.2 VAE Architecture Adaptation

The pretrained LTX-Video VAE is designed for three-channel natural RGB videos [1], whereas the physical trajectories considered in this thesis contain four channels: density, horizontal momentum, vertical momentum, and pressure. The VAE therefore requires adaptation at the observation interface.

Four trainable adaptation strategies are investigated: color-adapter adaptation, decoder-only fine-tuning, input-layer-and-decoder fine-tuning, and full-structure fine-tuning. In addition, the unfinetuned pretrained VAE is retained in the experimental study as a reference configuration, but is not considered an adaptation strategy. The four trainable strategies differ in how much of the pretrained representation is allowed to adapt to the physical-field domain.

4.2.1 Four-Channel Physical-Field Adaptation

The normalized physical trajectory is represented as

$$\tilde{\mathbf{X}} \in \mathbb{R}^{4 \times T \times H \times W}. \quad (4.7)$$

For the decoder-only, input-layer-and-decoder, and full-structure configurations, the native VAE input and output convolutions are expanded from three to four channels. The pretrained weights associated with the original three channels are retained, while weights corresponding to the newly introduced fourth channel are initialized to zero.

The internal latent dimensionality and spatiotemporal compression structure of the pretrained VAE are retained. In particular, the LTX-Video VAE uses strong spatiotemporal compression with spatial and temporal compression factors of 32 and 8, respectively [1]. The adaptation therefore modifies the mapping between physical observations and the latent representation without changing the dimensional structure expected by the pretrained latent generative model.

4.2.2 VAE Adaptation Strategies

Color-adapter adaptation. The color-adapter configuration retains the original three-channel VAE interface and introduces learned transformations before and after the VAE,

$$\tilde{\mathbf{X}}_4 \xrightarrow{\mathcal{A}_{\text{in}}} \mathbf{X}_3 \xrightarrow{\varepsilon_0} \mathbf{z} \xrightarrow{\mathcal{D}_\psi} \hat{\mathbf{X}}_3 \xrightarrow{\mathcal{A}_{\text{out}}} \hat{\tilde{\mathbf{X}}}_4. \quad (4.8)$$

Both adapters are point-wise $1 \times 1 \times 1$ convolutional MLPs with a 32-dimensional hidden representation and GELU activations. The input adapter maps four physical channels to three channels, while the output adapter maps the three decoder channels back to four physical channels. Their residual branches are initialized such that the adapters initially preserve the original three channels.

The pretrained encoder remains frozen in this configuration. The input and output adapters and the complete decoder are trainable. Because the adapters operate point-wise, they transform the channel representation without introducing additional spatial or temporal mixing. The decoder is nevertheless allowed to adapt to reconstruction of the physical-field domain.

Decoder-only fine-tuning. In the decoder-only configuration, the native VAE input and output layers are expanded to four channels, but the complete encoder remains frozen. Only the decoder is optimized. Because the newly introduced fourth-channel weights of the encoder input convolution are zero-initialized and subsequently frozen, the pressure channel does not contribute directly through the encoder input convolution in this configuration. The decoder must therefore reconstruct all four physical variables from a latent representation that contains no direct contribution from the pressure input channel. This constitutes an implementation-specific limitation of the decoder-only configuration investigated in this thesis and is taken into account when interpreting its experimental performance.

Input-layer-and-decoder fine-tuning. This configuration uses the same four-channel inflation of the native VAE interface as the decoder-only configuration, but additionally allows the encoder input convolution to adapt. Specifically, the expanded `conv_in` layer of the encoder and the complete decoder are trainable, whereas the remainder of the pretrained encoder trunk is frozen. The newly introduced fourth-channel weights of the encoder input convolution are therefore no longer fixed at zero. During fine-tuning, the input layer can learn a direct mapping from all four physical variables into the feature space expected by the frozen encoder trunk. The configuration can be summarized as

$$\tilde{\mathbf{X}}_4 \xrightarrow{\mathcal{E}_{\text{in},\phi}} \mathbf{h} \xrightarrow{\mathcal{E}_{\text{trunk},0}} \mathbf{z} \xrightarrow{\mathcal{D}_\psi} \hat{\mathbf{X}}_4, \quad (4.9)$$

where $\mathcal{E}_{\text{in},\phi}$ denotes the trainable expanded encoder input convolution and $\mathcal{E}_{\text{trunk},0}$ denotes the frozen remainder of the pretrained encoder.

This intermediate strategy separates two possible requirements of domain adaptation. Comparison with decoder-only fine-tuning tests whether allowing the fourth physical channel to enter the encoder through a learned input mapping is sufficient to improve reconstruction. Comparison with full-structure fine-tuning then tests whether adaptation must extend beyond the observation interface into the deeper encoder representation.

Full-structure fine-tuning. In the full-structure configuration, both the complete encoder and complete decoder are trainable,

$$\tilde{\mathbf{X}} \xrightarrow{\mathcal{E}_\phi} \mathbf{z} \xrightarrow{\mathcal{D}_\psi} \hat{\mathbf{X}}. \quad (4.10)$$

The zero-initialized fourth-channel input weights can therefore adapt during training together with the remainder of the encoder. This strategy provides the greatest flexibility among the investigated configurations for transferring the pretrained VAE to the four-channel physical domain. At the same time, because the encoder itself is optimized, this configuration also permits more substantial modification of the latent representation presented to the downstream DiT.

4.3 VAE Reconstruction Objectives

VAE adaptation is performed in the normalized physical-field representation. The reconstruction objective combines point-wise reconstruction accuracy with spatial-gradient preservation and, in selected experiments, structural similarity.

The padded temporal frames introduced for architectural compatibility are excluded from the reconstruction objective. All reconstruction losses are therefore evaluated only over the original 101 physical states.

4.3.1 Reconstruction Loss

Let

$$\mathbf{e} = \hat{\mathbf{X}} - \tilde{\mathbf{X}} \quad (4.11)$$

denote the reconstruction error over the original 101 physical states. The root-mean-square reconstruction loss is

$$\mathcal{L}_{\text{RMSE}} = \sqrt{\frac{1}{N} \sum_{i=1}^N e_i^2 + 10^{-8}}, \quad (4.12)$$

where

$$N = B \cdot 4 \cdot 101 \cdot H \cdot W. \quad (4.13)$$

The reduction is performed globally over the batch, physical channels, temporal states, and spatial dimensions. This loss directly penalizes point-wise differences in the normalized physical-field representation.

4.3.2 Gradient-Based Reconstruction Objective

To place additional emphasis on the reconstruction of sharp spatial structures, a gradient-based objective is introduced using forward finite differences in the two spatial directions.

Let δ_x and δ_y denote forward differences along the horizontal and vertical dimensions. The gradient-based loss is

$$\begin{aligned} \mathcal{L}_{\text{H1}} = & \frac{1}{N} \sum \left(\delta_x \hat{\mathbf{X}} - \delta_x \tilde{\mathbf{X}} \right)^2 \\ & + \frac{1}{N} \sum \left(\delta_y \hat{\mathbf{X}} - \delta_y \tilde{\mathbf{X}} \right)^2. \end{aligned} \quad (4.14)$$

The differences are padded with zeros at the final spatial row or column such that the reduction retains the original number of elements. No temporal derivative is included, and the differences are not scaled by the physical grid spacing.

The notation $\mathcal{L}_{\text{H1}}$ reflects the use of first-order spatial differences in addition to point-wise reconstruction. In this thesis, however, the term is interpreted specifically as a gradient-based or structure-sensitive reconstruction objective. It does not explicitly enforce the Euler equations, a PDE residual, or a conservation constraint and is therefore not treated as a physics-informed loss.

4.3.3 Structural Similarity Objective

Selected experiments additionally use the structural similarity index (SSIM) [23]. SSIM was originally proposed as an image-quality measure based on local structural information. Here, it is used as an

auxiliary reconstruction objective for the spatial structure of individual physical fields rather than as a perceptual image-quality metric.

Before SSIM evaluation, normalized values are mapped from $[-1, 1]$ to $[0, 1]$,

$$\mathbf{u} = \frac{\tilde{\mathbf{X}} + 1}{2}, \quad (4.15)$$

with the same transformation applied to the reconstruction.

Each physical channel of each temporal state is treated as an individual two-dimensional field. SSIM is evaluated using an 11×11 Gaussian window with standard deviation 1.5, data range 1, and constants

$$C_1 = 0.01^2, \quad C_2 = 0.03^2. \quad (4.16)$$

The SSIM loss is

$$\mathcal{L}_{\text{SSIM}} = 1 - \text{mean}(\text{SSIM}). \quad (4.17)$$

Only the corrected SSIM implementation described above is used for the controlled SSIM comparison reported in this thesis.

4.3.4 Combined VAE Objective

The reconstruction objective is expressed as

$$\mathcal{L}_{\text{rec}} = \lambda_{\text{R}}\mathcal{L}_{\text{RMSE}} + \lambda_{\text{H}}\mathcal{L}_{\text{H1}} + \lambda_{\text{S}}\mathcal{L}_{\text{SSIM}}. \quad (4.18)$$

The primary VAE configuration uses

$$\lambda_{\text{R}} = 1, \quad \lambda_{\text{H}} = 50, \quad \lambda_{\text{S}} = 0. \quad (4.19)$$

For the controlled SSIM ablation, $\lambda_{\text{S}} = 0.15$. The individual terms are combined using fixed weights. These coefficients are treated as experimental hyperparameters rather than physical weighting factors.

4.4 Physical Parameter Conditioning

In addition to the initial physical state, trajectory generation is conditioned on the ratio of specific heats γ . Since γ remains constant within each trajectory, it is represented as trajectory-level conditioning information rather than as an additional spatial channel.

4.4.1 Scalar Parameter Representation

The scalar parameter is first normalized according to

$$\tilde{\gamma} = \text{clip}\left(\frac{\gamma - 1}{2 - 1}, 0, 1\right). \quad (4.20)$$

The interval $[1, 2]$ is used as a fixed normalization range covering the γ values considered in the dataset. This representation keeps the global thermodynamic parameter separate from the spatially varying physical state.

4.4.2 Fourier-Feature Embedding

Directly providing a single scalar to a neural network can offer only a limited input representation. Following the general principle of Fourier-feature mappings for low-dimensional continuous inputs [21], the normalized scalar is therefore expanded using sine and cosine functions at multiple frequencies. The specific deterministic frequency schedule used here is an implementation choice of this thesis. The normalized scalar is mapped using 64 fixed frequencies,

$$f_k = 2^{4k/64}, \quad k = 0, \dots, 63. \quad (4.21)$$

For each frequency, sine and cosine components are evaluated,

$$\Phi(\tilde{\gamma}) = [\sin(2\pi f_k \tilde{\gamma}), \cos(2\pi f_k \tilde{\gamma})]_{k=0}^{63}, \quad (4.22)$$

resulting in a 128-dimensional representation. This representation is processed by an MLP,

$$128 \rightarrow 256 \rightarrow 256 \rightarrow 4 \cdot 4096, \quad (4.23)$$

and reshaped into

$$\mathbf{C}_\gamma \in \mathbb{R}^{4 \times 4096}. \quad (4.24)$$

The MLP uses Layer Normalization, GELU activations, and dropout with probability 0.1. These architectural choices follow standard neural-network components; the dimensions and number of output tokens are selected to match the conditioning interface used in the adapted model.

4.4.3 Conditioning Injection

The four scalar-conditioning tokens replace the text-conditioning representation of the pretrained video model. They are supplied as encoder hidden states through the existing caption-projection and cross-attention conditioning pathway of the DiT.

Thus, the DiT receives two complementary forms of conditioning. The initial latent frame provides spatial information about the specific initial state, while $\mathbf{C}_\gamma$ provides trajectory-level thermodynamic conditioning. The scalar condition is used during both training and inference. No conditioning dropout or classifier-free guidance is applied. Consequently, inference uses the conditional prediction directly, corresponding to a guidance scale of one. “

“`latex`

4.5 Latent Representation and DiT Fine-Tuning

After VAE adaptation, a selected encoder defines the latent representation on which the DiT is fine-tuned. The VAE is fixed during this stage, and latent trajectories are precomputed before DiT training. This preserves the two-stage training procedure introduced in section 4.1.3 and ensures that each DiT experiment is associated with a fixed VAE representation.

4.5.1 VAE Latent Representation

The pretrained LTX-Video VAE performs strong temporal and spatial compression, with temporal and spatial compression factors of 8 and 32, respectively [1]. Each physical trajectory in the present dataset contains 101 states. To obtain a sequence length compatible with the temporal compression structure of the VAE, the final physical state is repeated four times,

$$\tilde{\mathbf{X}}_{0:100} \longrightarrow \tilde{\mathbf{X}}_{0:104}. \quad (4.25)$$

The resulting 105-state sequence is encoded into 14 latent frames,

$$105 = 1 + 13 \times 8. \quad (4.26)$$

The first latent frame is associated with the initial physical state, while the remaining 13 latent frames represent temporally compressed portions of the subsequent trajectory. The four repeated states are introduced only to satisfy the temporal structure required by the encoder; they are not treated as additional physical observations.

The spatial compression factor is 32 [1]. Consequently,

$$256 \times 256 \rightarrow 8 \times 8, \quad (4.27)$$

while

$$128 \times 128 \rightarrow 4 \times 4. \quad (4.28)$$

The encoder parameterizes a Gaussian latent posterior, consistent with the variational-autoencoder formulation [5]. In the present pipeline, the posterior mean produced by the encoder is used as the deterministic latent representation; no posterior sampling is performed during VAE reconstruction, latent precomputation, or DiT fine-tuning.

Before DiT processing, the posterior mean is transformed using the latent normalization convention of the pretrained LTX-Video model,

$$\mathbf{z} = \frac{(\boldsymbol{\mu}_{\text{enc}} - \mathbf{m})s}{\boldsymbol{\sigma}_{\text{lat}}}, \quad (4.29)$$

where $\mathbf{m}$ and $\boldsymbol{\sigma}_{\text{lat}}$ denote the pretrained per-channel latent statistics and s denotes the VAE scaling factor. The corresponding inverse transformation is applied before decoding.

Importantly, these pretrained normalization constants are retained after VAE adaptation. Thus, although the encoder parameters may change during fine-tuning, its output is transformed through the same normalization interface expected by the downstream latent generative model.

4.5.2 First-Frame Conditioning

Only the first latent frame is used as the spatial conditioning frame. During training, its tokens remain equal to their clean latent values and their flow-matching timestep is set to zero. These conditioning tokens are excluded from the training loss.

Let $\mathcal{M}$ denote the set of non-conditioning tokens corresponding to latent frames $1, \dots, 13$. The conditional generative problem is therefore

$$p_{\theta}(\mathbf{z}_{1:13} \mid \mathbf{z}_0, \gamma). \quad (4.30)$$

The future latent frames are modeled jointly rather than through repeated autoregressive transitions. Consequently, the model learns a conditional distribution over the complete compressed future trajectory given the initial latent state and the physical parameter γ .

4.5.3 Flow-Matching Objective

DiT fine-tuning follows the flow-matching formulation introduced in section 2.2.4 [8]. For a clean latent sequence $\mathbf{z}$, Gaussian noise

$$\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \quad (4.31)$$

is sampled together with a flow parameter σ . For the non-conditioning tokens, the intermediate latent state is constructed as

$$\mathbf{z}_{\sigma} = (1 - \sigma)\mathbf{z} + \sigma\epsilon, \quad (4.32)$$

while the first-frame conditioning tokens are replaced by their clean latent values. For this linear interpolation, the target velocity is

$$\mathbf{v}^* = \epsilon - \mathbf{z}. \quad (4.33)$$

The DiT predicts

$$\mathbf{v}_{\theta} = \mathbf{v}_{\theta}(\mathbf{z}_{\sigma}, \sigma; \mathbf{z}_0, \mathbf{C}_{\gamma}). \quad (4.34)$$

The flow parameter follows the shifted logit-normal sampling strategy used by the training implementation. The corresponding timestep is supplied to the transformer, while conditioned first-frame tokens receive timestep zero.

The training objective is a masked mean-squared error over the non-conditioning tokens,

$$\mathcal{L}_{\text{FM}} = \frac{1}{|\mathcal{M}|D} \sum_{i \in \mathcal{M}} \sum_{d=1}^D \left(v_{\theta, i, d} - v_{i, d}^* \right)^2, \quad (4.35)$$

where D denotes the feature dimension associated with each token.

No decoded-space reconstruction loss, physical-field VRMSE objective, or additional latent-alignment loss is used during DiT fine-tuning. The DiT is optimized solely through the masked latent-space flow-matching objective. This distinction is important because the training objective evaluates latent velocity prediction, whereas the final surrogate is evaluated after decoding the generated latents into physical fields.

4.5.4 Conditional Trajectory Rollout

At inference time, the model receives only the normalized initial physical state $\tilde{\mathbf{X}}_0$ and the scalar parameter γ . The initial state is encoded into $\mathbf{z}_0$, while γ is transformed into $\mathbf{C}_\gamma$ using the conditioning module described in section 4.4.

Conditioned on these quantities, the DiT jointly generates the 13 future latent frames,

$$\hat{\mathbf{z}}_{1:13} \sim p_\theta(\mathbf{z}_{1:13} \mid \mathbf{z}_0, \mathbf{C}_\gamma). \quad (4.36)$$

The flow trajectory is numerically integrated from the noise endpoint toward the data endpoint using the inference procedure of the adapted latent generative model. The resulting latent sequence consists of the conditioned first frame and 13 generated future latent frames.

The complete 14-frame latent trajectory is then transformed back through the inverse latent normalization and decoded into 105 physical-field states,

$$\hat{\tilde{\mathbf{X}}}_{0:104} = \mathcal{D}_\psi(\mathbf{z}_0, \hat{\mathbf{z}}_{1:13}). \quad (4.37)$$

The final four padding states are discarded, leaving

$$\hat{\tilde{\mathbf{X}}}_{0:100}. \quad (4.38)$$

The decoded first state is a VAE reconstruction of the supplied initial state rather than an exact copy of the input. End-to-end performance is therefore evaluated on the complete decoded trajectory in normalized physical-field space.

4.6 VAE–DiT Compatibility

Although VAE and DiT adaptation are performed in separate stages, the two components remain functionally coupled. The VAE determines both the latent representation on which the DiT is trained and the decoding transformation through which generated latents are mapped back to physical fields. This interaction is especially relevant in latent generative architectures such as LTX-Video, where the VAE and transformer jointly define the complete observation-to-latent-to-observation generation pipeline [1]. This motivates the concept of VAE–DiT compatibility used in this thesis. Compatibility is treated primarily as a functional downstream property rather than as similarity to the original pretrained latent representation.

4.6.1 Motivation

Successful VAE reconstruction requires

$$\mathcal{D}_\psi(\mathcal{E}_\phi(\tilde{\mathbf{X}})) \approx \tilde{\mathbf{X}}. \quad (4.39)$$

During conditional trajectory generation, however, the decoder receives latent states generated by the DiT,

$$\mathcal{D}_\psi(\mathbf{z}_0, \hat{\mathbf{z}}_{1:L}) \approx \tilde{\mathbf{X}}_{0:T}. \quad (4.40)$$

These two conditions are not equivalent. Accurate VAE reconstruction demonstrates that the adapted encoder and decoder form an effective autoencoding pair, but does not by itself establish that the corresponding latent representation is equally suitable for downstream latent generative modeling.

This distinction is particularly important because the DiT does not generate exact encoder outputs. Even a well-trained latent generative model produces predicted latent states with residual errors relative to the encoded reference trajectory. The final physical-field accuracy therefore depends both on latent generation and on how the adapted decoder maps such generated latents back to the physical-field representation.

4.6.2 Representation Adaptation

Let $\mathcal{E}_0$ denote the pretrained encoder and $\mathcal{E}_\phi$ the adapted encoder. For the same physical trajectory,

$$\mathbf{z}^{(0)} = \mathcal{E}_0(\tilde{\mathbf{X}}), \quad \mathbf{z}^{(\phi)} = \mathcal{E}_\phi(\tilde{\mathbf{X}}). \quad (4.41)$$

Fine-tuning the encoder can therefore modify the latent representation relative to that produced by the pretrained model. Such representation adaptation is not inherently undesirable, since the target physical domain differs substantially from natural RGB video. Nevertheless, it provides a possible mechanism through which VAE adaptation may affect subsequent DiT fine-tuning.

Latent-space diagnostics were used during model development, but representation shift is not treated as the primary quantitative evaluation criterion in the controlled experiments of this thesis. In particular, downstream performance is not inferred from a single geometric distance between pretrained and adapted latent representations.

This distinction avoids assuming that the original pretrained representation is necessarily optimal for the physical domain. The relevant question is instead whether the representation obtained after adaptation can support accurate downstream physical-field trajectory generation.

4.6.3 Functional Compatibility Evaluation

VAE–DiT compatibility is therefore evaluated through the complete downstream pipeline,

$$\begin{aligned} &\text{VAE adaptation} \rightarrow \text{trajectory encoding} \rightarrow \text{DiT fine-tuning} \\ &\quad \rightarrow \text{conditional rollout} \rightarrow \text{VAE decoding} \rightarrow \text{normalized physical-field evaluation.} \end{aligned}$$

(4.42)

This functional definition does not assume that similarity to the original pretrained representation is itself the desired objective. A representation may change substantially while remaining suitable for downstream adaptation. Conversely, strong reconstruction performance does not necessarily imply strong decoded rollout performance.

Accordingly, VAE reconstruction and downstream trajectory generation are reported as separate evaluation levels. The corresponding metrics and controlled experimental protocol are introduced in chapter 5, and the observed relationship between VAE reconstruction and downstream DiT performance is analyzed in chapter 6.

4.7 Compatibility-Preserving VAE Regularization

The compatibility considerations above motivate the investigation of additional regularization during VAE adaptation. The reconstruction objective directly optimizes the autoencoding of physical trajectories but does not explicitly incorporate the downstream latent generative model.

Two additional strategies are therefore investigated: KL regularization and DiT-guided distillation. The first introduces a distributional constraint on the VAE posterior, while the second uses the frozen pre-trained DiT as an auxiliary teacher. These approaches are evaluated empirically rather than assumed to preserve downstream compatibility.

4.7.1 KL Regularization

Variational autoencoders represent the latent state through an approximate posterior distribution and conventionally regularize this distribution toward a prior using the Kullback–Leibler divergence [5]. The VAE encoder used here parameterizes a diagonal Gaussian posterior with mean μ and log variance $\log \sigma^2$.

For a standard normal prior, the KL term used in this work is

$$\mathcal{L}_{\text{KL}} = \frac{1}{B} \sum_{b=1}^B \frac{1}{2} \sum_{c,t,h,w} (\mu^2 + \exp(\log \sigma^2) - \log \sigma^2 - 1). \quad (4.43)$$

The KL term is evaluated on the raw encoder posterior before the pretrained latent normalization and covers the complete encoded latent sequence, including the temporally padded portion.

The corresponding training objective is

$$\mathcal{L}_{\text{VAE+KL}} = \mathcal{L}_{\text{rec}} + \lambda_{\text{KL}} \mathcal{L}_{\text{KL}}. \quad (4.44)$$

Unlike the conventional stochastic VAE training formulation [5], the reconstruction pathway in the present implementation uses the posterior mean rather than samples obtained through reparameterization. The KL term is therefore introduced specifically as a distributional regularizer during representation adaptation.

It should not be interpreted as explicitly forcing the adapted representation to reproduce the original pretrained latent space. Rather, it constrains the Gaussian posterior toward the standard-normal prior while the reconstruction objective adapts the encoder and decoder to the physical-field domain.

4.7.2 DiT-Guided Distillation

The second strategy introduces an auxiliary signal from the stock pretrained LTX-Video DiT during VAE adaptation. The general motivation is related to score distillation, in which a frozen pretrained diffusion model provides a generative-prior signal for optimizing another differentiable representation [18]. In the present work, this principle is adapted to the flow-matching parameterization and latent representation of LTX-Video.

The resulting objective is therefore referred to as an *SDS-based* objective rather than an exact reproduction of the original Score Distillation Sampling formulation. The frozen stock LTX-Video DiT acts as the teacher, while the trainable VAE encoder receives the resulting latent-space gradient.

Let

$$\mathbf{z}_\theta = \mathcal{E}_\theta(\tilde{\mathbf{X}}) \quad (4.45)$$

denote the normalized latent representation obtained from the posterior mean of the current VAE encoder. For the teacher forward pass, the latent is detached from the computational graph and perturbed according to

$$\mathbf{z}_\sigma = (1 - \sigma) \text{sg}(\mathbf{z}_\theta) + \sigma \epsilon, \quad (4.46)$$

where $\text{sg}(\cdot)$ denotes the stop-gradient operation and

$$\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (4.47)$$

As in DiT fine-tuning, the first latent frame is retained as a clean conditioning frame and excluded from the distillation loss. The frozen teacher predicts

$$\hat{\mathbf{v}}_\phi = \text{sg}[\mathbf{v}_\phi(\mathbf{z}_\sigma, \sigma, \mathbf{c}_\gamma)], \quad (4.48)$$

where ϕ denotes the parameters of the frozen stock LTX-Video DiT.

The conditioning tokens $\mathbf{c}_\gamma$ depend deterministically on the trajectory's γ value and are supplied through the teacher's existing conditioning pathway. However, the scalar-embedding module used for this teacher signal is randomly initialized and frozen rather than learned for the physical domain. Consequently, these tokens should not be interpreted as physically learned conditioning of the pretrained teacher.

For the linear flow parameterization used in this work, the corresponding velocity target is

$$\mathbf{v}_\theta^* = \epsilon - \mathbf{z}_\theta. \quad (4.49)$$

The implemented SDS-based distillation objective is

$$\mathcal{L}_{\text{SDS}} = \frac{1}{|\mathcal{M}|D} \sum_{i \in \mathcal{M}} \sum_{d=1}^D [\hat{\mathbf{v}}_{\phi,i,d} - (\epsilon_{i,d} - \mathbf{z}_{\theta,i,d})]^2, \quad (4.50)$$

where $\mathcal{M}$ denotes the non-conditioning latent tokens.

The gradient structure is important. Both the noisy teacher input and the teacher prediction are detached from the computational graph. Consequently, the gradient reaches the VAE only through $\mathbf{z}_\theta$ in the target velocity. Up to the positive scalar factor introduced by the mean-squared-error reduction,

$$\nabla_{\mathbf{z}_\theta} \mathcal{L}_{\text{SDS}} \propto \mathbf{z}_\theta - (\epsilon - \hat{\mathbf{v}}_\phi). \quad (4.51)$$

The distillation term therefore affects the trainable encoder parameters but does not directly propagate through the decoder or through the frozen teacher.

The complete VAE objective becomes

$$\mathcal{L}_{\text{VAE+SDS}} = \mathcal{L}_{\text{rec}} + \lambda_{\text{SDS}} \mathcal{L}_{\text{SDS}}. \quad (4.52)$$

This procedure is referred to as *DiT-guided distillation* because the pretrained generative model provides an additional latent-space training signal during VAE adaptation. It should not be interpreted as guidance from a CFD-adapted teacher: the teacher is the frozen stock video DiT and has not been fine-tuned on the Euler-flow dataset.

4.7.3 Reconstruction–Compatibility Trade-off

KL regularization and DiT-guided distillation introduce objectives beyond direct reconstruction of the target physical fields. Their effects are therefore evaluated at two distinct levels.

At the VAE level, reconstruction performance evaluates

$$\mathcal{D}_\psi(\mathcal{E}_\phi(\tilde{\mathbf{X}})) \approx \tilde{\mathbf{X}}. \quad (4.53)$$

At the complete-pipeline level, the relevant condition is

$$\mathcal{D}_\psi(\mathbf{z}_0, \hat{\mathbf{z}}_{1:L}) \approx \tilde{\mathbf{X}}_{0:T}. \quad (4.54)$$

The two evaluation levels need not respond identically to regularization. Stronger regularization may constrain representation adaptation while simultaneously degrading reconstruction of the target physical domain. Conversely, minimizing reconstruction error alone does not directly optimize the representation for downstream latent generative modeling.

The regularization experiments are therefore interpreted through both VAE reconstruction and downstream decoded trajectory performance. This allows the analysis to distinguish reconstruction quality from functional VAE–DiT compatibility without requiring compatibility to be represented by a single latent-space distance metric. “

Chapter 5

Experimental Setup

This chapter specifies the experimental protocol used to evaluate the adaptation framework introduced in chapter 4. The physical problem, dataset, trajectory representation, normalization, and training–validation–test partition are defined in chapter 3; the present chapter therefore focuses on the implementation choices required to train and evaluate the VAE and DiT stages consistently.

The experimental study is organized around two levels of evaluation. First, the VAE is evaluated independently to determine how adaptation depth, reconstruction objective, learning rate, and spatial resolution affect physical-field reconstruction. Second, selected VAE configurations are used to construct latent training representations for separate DiT models, allowing the complete VAE–DiT pipeline to be evaluated in decoded physical space.

Unless otherwise stated, experiments follow a common reference configuration and modify only the component under investigation. This controlled protocol provides the basis for the comparisons presented in chapter 6.

5.1 Data Processing for Model Training

All experiments use the Euler Multi-Quadrants dataset representation and trajectory-level partition defined in chapter 3. The primary experiments operate on the 256×256 representation, while the 128×128 representation is used only for the dedicated spatial-resolution ablation. The corresponding resolution-specific normalization statistics defined in chapter 3 are applied before model processing. The pretrained LTX-Video VAE imposes an additional temporal requirement on the input sequence. Each physical trajectory contains 101 states and is therefore padded to 105 frames by repeating its final state four times. These additional frames are architectural padding only. During VAE training and reconstruction evaluation, losses and metrics are restricted to the original 101 physical states.

For DiT training, the padded trajectories are encoded using the VAE checkpoint associated with the corresponding experiment. A 105-frame physical trajectory produces 14 latent frames. The first latent frame represents the initial-state condition, while the remaining 13 latent frames constitute the generative portion of the trajectory.

Latent trajectories are precomputed after VAE adaptation and reused during DiT fine-tuning. The VAE is therefore fixed during the downstream DiT stage. At inference time, only the initial physical state is encoded and supplied as the spatial condition; future ground-truth physical states are not provided to the generative model.

The model-specific processing pipeline can thus be summarized as

$$\begin{aligned} \text{normalized physical trajectory} &\rightarrow \text{temporal padding} \rightarrow \text{VAE encoding} \\ &\rightarrow \text{latent trajectory} \rightarrow \text{DiT training or generation.} \end{aligned} \tag{5.1}$$

After generation, the predicted latent trajectory is decoded using the VAE paired with that DiT. The resulting 105-frame physical sequence is truncated to the original 101-state horizon before quantitative evaluation.

5.2 Training Configuration

The VAE and DiT are trained sequentially following the two-stage adaptation strategy described in chapter 4. Unless otherwise stated, the optimization settings within each stage are held fixed across controlled comparisons.

5.2.1 VAE Training

The primary VAE experiments use the 256×256 training representation defined in chapter 3. Training is performed for 20 epochs with an effective batch size of 32 distributed over eight XPU processes. A fixed random seed of 42 is used for the controlled experiments.

The reference configuration uses full-structure fine-tuning, in which both the encoder and decoder are trainable. The channel adaptation and parameter initialization are described in section 4.2.

The initial learning rate of the reference configuration is

$$\eta_0 = 5 \times 10^{-5}. \quad (5.2)$$

Training begins with a 200-step linear warm-up using an initial warm-up factor of 0.01. The learning rate is subsequently reduced using a cosine schedule to

$$\eta_{\min} = 10^{-6}. \quad (5.3)$$

No weight decay is applied during VAE fine-tuning.

The reference reconstruction objective is

$$\mathcal{L}_{\text{rec}} = \mathcal{L}_{\text{RMSE}} + 50\mathcal{L}_{\text{H1}}, \quad (5.4)$$

with the individual objectives defined in section 4.3. SSIM is disabled in the reference configuration. The alternative reconstruction objectives evaluated in the controlled loss ablation are specified in section 5.4.2.

All reconstruction objectives operate on the normalized physical-field representation defined in chapter 3. Although trajectories are internally padded from 101 to 105 frames, the additional padding frames are excluded before the reconstruction objective is evaluated.

During VAE training, checkpoint selection is based on the pooled validation VRMSE, denoted as `val_vrmse`.

The channel-mean validation VRMSE, `val_vrmse_chmean`, is reported separately and is used as the primary scientific metric for comparing reconstruction quality across the four physical variables.

For the compatibility-oriented experiments, the same reference reconstruction configuration is augmented by either KL regularization or DiT-guided distillation as defined in section 4.6. The corresponding regularization coefficients are specified as part of the controlled ablation protocol in section 5.4.5.

5.2.2 DiT Training

DiT fine-tuning is performed only after the corresponding VAE adaptation stage has been completed. For each selected VAE checkpoint, the encoder is fixed and used to construct the latent training representation. The VAE is not jointly optimized with the DiT during this stage.

All downstream DiT experiments use the same optimization configuration. The complete transformer together with the scalar γ embedding module is fine-tuned using AdamW [9] with an initial learning rate

$$\eta_{\text{DiT}} = 10^{-5}. \quad (5.5)$$

The optimizer uses

$$\beta_1 = 0.9, \quad \beta_2 = 0.999, \quad \epsilon = 10^{-8}, \quad (5.6)$$

with a weight decay of 0.01. Training is performed for 8000 optimization steps. With an effective batch size of 32 and 4000 training trajectories, this corresponds to approximately 64 effective passes over the training set, although the implementation is step-based rather than epoch-based.

A cosine learning-rate scheduler is used with

$$T_{\text{max}} = 8000, \quad \eta_{\text{min}} = 10^{-8}, \quad (5.7)$$

without an additional warm-up stage. Gradients are clipped to a maximum norm of 1.0.

Training uses eight XPU processes with a per-process batch size of one and four gradient-accumulation steps, resulting in

$$B_{\text{eff}} = 8 \times 1 \times 4 = 32. \quad (5.8)$$

DiT fine-tuning uses BF16 mixed precision, and the random seed is fixed to 42. These optimization settings are identical across all downstream comparisons; the VAE used to construct the latent training representation is the principal component varied between experiments.

Each trajectory contains 14 latent frames after VAE encoding. First-frame conditioning is applied with probability

$$p_{\text{FFC}} = 1, \quad (5.9)$$

so that every training sample is conditioned on its initial latent frame. The conditioned tokens remain clean, receive a flow-matching timestep of zero, and are excluded from the training objective.

The remaining latent tokens are optimized using the masked flow-matching objective defined in section 4.5.3. No decoded physical-field reconstruction loss is included during DiT fine-tuning.

Each trajectory is additionally conditioned on its scalar γ value. The scalar is transformed into four conditioning tokens and supplied through the transformer's conditioning pathway as described in section 4.4. No classifier-free guidance or conditioning dropout is used.

At evaluation time, generation starts from the encoded initial physical state and the corresponding γ value. The 13 future latent frames are generated jointly rather than autoregressively. The evaluation configuration uses 20 flow-integration steps and a guidance scale of

$$s_{\text{guidance}} = 1. \quad (5.10)$$

Each DiT is evaluated together with the VAE checkpoint that defines its training representation. This pairing is preserved throughout the downstream study so that decoded trajectory performance characterizes the complete VAE–DiT pipeline rather than the DiT in isolation.

The downstream comparisons include the unfinetuned and full-finetuned VAE representations together with the retained KL- and SDS-regularized VAE configurations. Because each independently adapted VAE defines a different latent representation and scale, DiT training or validation losses are not used to rank models across VAE configurations. Cross-configuration comparison is instead performed after decoding using the physical-field metric defined in section 5.3.3.

5.3 Evaluation Metrics

The evaluation protocol distinguishes three quantities: VAE reconstruction quality, latent-space DiT prediction error, and end-to-end decoded trajectory quality. This distinction is necessary because accurate VAE reconstruction or small latent prediction error does not by itself establish accurate decoded physical-field generation.

Unless otherwise stated, physical-field metrics are computed in the normalized and clipped representation defined in chapter 3. They therefore quantify error in normalized physical-field space rather than in the original dimensional physical units.

5.3.1 Physical-Field VRMSE

Let s denote a trajectory sample and c one of the four physical channels. The channel-wise variance-normalized root-mean-square error is defined as

$$\text{VRMSE}_{s,c} = \sqrt{\frac{\text{mean}_{t,h,w} \left[\left(\hat{x}_{s,c,t,h,w} - x_{s,c,t,h,w} \right)^2 \right]}{\text{var}_{t,h,w} \left(x_{s,c,t,h,w} \right) + 10^{-8}}}. \quad (5.11)$$

The variance is the population variance. The channel-mean metric is then

$$\text{VRMSE}_{\text{chmean}} = \frac{1}{N} \sum_{s=1}^N \left(\frac{1}{4} \sum_{c=1}^4 \text{VRMSE}_{s,c} \right). \quad (5.12)$$

This formulation normalizes the error of each physical variable by its own spatiotemporal variance before averaging across channels. It therefore prevents variables with larger numerical variance from dominating the aggregate solely because of their scale and is used as the primary physical-field reporting metric. A pooled variant is additionally computed. For each trajectory,

$$\text{VRMSE}_{s,\text{pooled}} = \sqrt{\frac{\text{mean}_{c,t,h,w} \left[\left(\hat{x}_{s,c,t,h,w} - x_{s,c,t,h,w} \right)^2 \right]}{\text{var}_{c,t,h,w} \left(x_{s,c,t,h,w} \right) + 10^{-8}}}, \quad (5.13)$$

followed by

$$\text{VRMSE}_{\text{pooled}} = \frac{1}{N} \sum_{s=1}^N \text{VRMSE}_{s,\text{pooled}}. \quad (5.14)$$

For VAE reconstruction, these metrics are computed over the original 101 physical states, excluding temporal padding. The pooled validation metric corresponds to `val_vrmse` and is used for checkpoint selection, whereas `val_vrmse_chmean` is emphasized for scientific comparison. Per-channel VRMSE values are additionally reported to identify differences among density, the two momentum components, and pressure.

Validation metrics are used during development and checkpoint selection. Final quantitative comparisons in chapter 6 are computed on the independent 500-trajectory test partition defined in chapter 3.

5.3.2 Latent-Space DiT Error

Latent-space VRMSE is retained as a diagnostic measure of DiT prediction quality. For an evaluated trajectory, predicted latent values are compared against the corresponding normalized reference latent representation generated by the same VAE checkpoint,

$$\text{VRMSE}_s^{\text{latent}} = \sqrt{\frac{\text{mean}[(\hat{\mathbf{z}}_s - \mathbf{z}_s)^2]}{\text{var}(\mathbf{z}_s) + 10^{-8}}}. \quad (5.15)$$

The reduction jointly covers the latent-channel, temporal, and spatial dimensions included by the evaluation implementation, and the reported value is obtained by averaging over evaluated trajectories.

Latent VRMSE is treated strictly as a diagnostic rather than as the primary downstream metric. In particular, it is not interpreted as a quantitative measure of representation shift between different VAEs. Because each VAE defines its own latent representation, this metric characterizes prediction error only within the latent space associated with a particular VAE checkpoint.

5.3.3 End-to-End Trajectory Error

The primary downstream evaluation measures the complete VAE–DiT pipeline after the generated latent trajectory has been decoded. The physical-field VRMSE definitions above are applied to the decoded prediction and corresponding normalized reference trajectory.

The primary end-to-end metric is

$$\text{VRMSE}_{\text{chmean}}^{\text{VAE+DiT}} = \frac{1}{N} \sum_{s=1}^N \left(\frac{1}{4} \sum_{c=1}^4 \text{VRMSE}_{s,c}^{\text{VAE+DiT}} \right). \quad (5.16)$$

The complete decoded trajectory is evaluated over the original 101-state horizon. This includes state zero, which is reconstructed through the VAE rather than copied directly from the input. Consequently, the end-to-end metric reflects both VAE reconstruction behavior and the quality of the future latent trajectory generated by the DiT.

The corresponding pooled metric is retained as a secondary measure. For downstream comparisons, however, the channel-mean decoded VRMSE is given priority because it evaluates all four normalized physical variables after completion of the full VAE–DiT generation and decoding pipeline.

The evaluation hierarchy is therefore

VAE reconstruction:	$\text{VRMSE}_{\text{chmean}}$	(5.17)
DiT latent diagnostic:	$\text{VRMSE}_{\text{latent}}$	
end-to-end generation:	$\text{VRMSE}_{\text{chmean}}^{\text{VAE+DiT}}$	

5.4 Ablation Protocol and Experimental Controls

The experimental study is organized as a sequence of controlled ablations around a common VAE reference configuration. The reference uses full-structure fine-tuning at 256×256 resolution, an initial learning rate of 5×10^{-5} , and the $\text{RMSE}-H^1$ reconstruction objective defined in eq. (5.4).

Unless the corresponding quantity is itself under investigation, the dataset partition, preprocessing, training duration, optimization configuration, and evaluation protocol are retained.

5.4.1 VAE Adaptation Strategy

The first ablation investigates how the degree of VAE adaptation affects reconstruction of the physical-field trajectories. Four trainable adaptation strategies are compared:

1. color-adapter adaptation,
2. strict decoder-only fine-tuning,
3. input-layer-and-decoder fine-tuning, and
4. full-structure fine-tuning.

The architectures of these configurations are defined in section 4.2. They form a progressive comparison of how much of the pretrained representation is allowed to adapt to the four-channel physical-field domain.

The strict decoder-only configuration retains the zero-initialized fourth-channel encoder-input weights in a frozen state. Consequently, pressure cannot contribute directly through the encoder input convolution. The input-layer-and-decoder configuration additionally makes the expanded `encoder.conv_in` trainable, allowing this input-interface limitation to be separated from adaptation of the deeper encoder representation.

An unfinetuned four-channel pretrained VAE is additionally evaluated as a reference. It is not treated as a trainable adaptation strategy but provides a common baseline for quantifying the effect of VAE fine-tuning and is also retained for the downstream VAE-DiT comparison.

5.4.2 Reconstruction-Objective Ablation

The reconstruction-objective ablation compares

$$\mathcal{L}_{\text{RMSE}}, \tag{5.18}$$

$$\mathcal{L}_{\text{RMSE}} + 50\mathcal{L}_{H^1}, \tag{5.19}$$

and

$$\mathcal{L}_{\text{RMSE}} + 50\mathcal{L}_{H^1} + 0.15\mathcal{L}_{\text{SSIM}}. \tag{5.20}$$

The $\text{RMSE}-H^1$ configuration serves as the reference. The comparison with RMSE alone evaluates the contribution of the spatial-gradient term, while the third configuration evaluates the additional SSIM objective.

Only experiments using the corrected SSIM implementation are included in the controlled comparison. Earlier development runs using the previous SSIM implementation are excluded from the final analysis.

5.4.3 Learning-Rate Sensitivity

Full-structure VAE fine-tuning is quantitatively compared using

$$\eta_0 \in \{10^{-5}, 5 \times 10^{-5}\}. \quad (5.21)$$

All other components of the reference training configuration are retained. A preliminary run using an initial learning rate of 10^{-4} became numerically unstable and is therefore excluded from the final quantitative comparison.

5.4.4 Spatial-Resolution Ablation

The reference VAE configuration is trained using the two dataset resolutions defined in chapter 3,

$$256 \times 256 \quad \text{and} \quad 128 \times 128. \quad (5.22)$$

The corresponding latent spatial resolutions are 8×8 and 4×4 . The trajectory-level partition and remaining training protocol are unchanged, while each resolution uses its corresponding normalization statistics.

For the final comparison, the VAE trained on the 128×128 representation is applied directly to the same 256×256 test trajectories used for the reference model. No spatial resizing, interpolation, downsampling, or upsampling is performed during evaluation. Because the LTX-Video VAE is fully convolutional, the lower-resolution-trained model can directly encode and decode the 256×256 input.

The 128×128 -trained model retains the normalization statistics associated with its training representation. These statistics differ from the corresponding 256×256 statistics by less than 0.2%.

This experiment therefore evaluates whether a VAE adapted using lower-resolution trajectories transfers directly to a finer spatial discretization. It should not be interpreted as a low-resolution reconstruction followed by upsampling.

The resolution comparison is restricted to VAE reconstruction; the 128×128 configuration is not used as a separate downstream DiT setting.

5.4.5 Compatibility-Oriented Regularization

The final group of VAE experiments investigates whether the latent-space regularization strategies introduced in section 4.6 affect both reconstruction and downstream VAE–DiT performance.

KL regularization is evaluated using

$$\lambda_{\text{KL}} \in \{10^{-7}, 10^{-5}\}, \quad (5.23)$$

and DiT-guided distillation using

$$\lambda_{\text{SDS}} \in \{10^{-4}, 10^{-2}, 1\}. \quad (5.24)$$

A run with $\lambda_{\text{KL}} = 10^{-6}$ became numerically unstable and is excluded from the final quantitative comparison. An earlier development configuration using $\lambda_{\text{SDS}} = 0.1$ is likewise not included in the final controlled sweep.

Each retained configuration is evaluated at two levels. First, the adapted VAE is evaluated independently using the reconstruction metrics defined in section 5.3. Second, its latent representation is used to train a separate DiT under the common downstream protocol. The resulting trajectory is decoded using the corresponding VAE and evaluated in physical space.

This two-level protocol allows the effect of latent regularization on reconstruction and downstream generation to be compared without assuming that the two quantities must follow the same ordering.

5.4.6 Run-to-Run Variability

To estimate the magnitude of stochastic training variation, the primary 256×256 VAE baseline is repeated with additional random seeds while retaining the same dataset partition, architecture, reconstruction objective, learning-rate schedule, batch size, and training duration.

The resulting spread in test-set $\text{VRMSE}_{\text{chmean}}$ provides an empirical reference for interpreting small differences between VAE ablation arms. This analysis is reported separately from the single-seed controlled ablations and does not alter the reference configuration used elsewhere.

5.4.7 Experimental Controls

Across the controlled VAE experiments, the trajectory-level partition, preprocessing, normalization, temporal padding, and evaluation procedure are held fixed. The primary controlled runs use a random seed of 42; only the dedicated run-to-run variability experiment changes the training seed.

Except where explicitly ablated, VAE experiments use 256×256 inputs, full-structure fine-tuning, 20 training epochs, an effective batch size of 32, an initial learning rate of 5×10^{-5} , and the $\text{RMSE}-H^1$ reconstruction objective.

For downstream experiments, each DiT is paired with the VAE checkpoint used to construct its latent training representation. The VAE remains fixed during DiT fine-tuning, while the DiT optimization configuration is identical across comparisons.

VAE reconstruction and downstream generation are evaluated separately. The experimental protocol therefore does not assume that the VAE configuration with the lowest reconstruction error must also produce the lowest decoded trajectory error; their relationship is examined empirically in chapter 6.

5.5 Computational Resources

The experiments were conducted on the `sng_pvc` partition of the HPC system. Each accelerator node provides four Intel Data Center GPU Max 1550 accelerators, with two computational tiles per GPU, resulting in eight XPU tiles per node.

Both VAE and DiT training use one compute node and all eight XPU tiles. Distributed training is launched through SLURM and `accelerate` using eight processes. Communication between XPU processes is handled through `oneCCL`, and the implementation uses the PyTorch XPU backend.

For VAE training, model parameters are retained in FP32 while forward computation is performed under BF16 automatic mixed precision. DiT fine-tuning is likewise performed using BF16 mixed precision.

Latent preprocessing and DiT evaluation are executed as single-process jobs using one XPU tile. Latent trajectories are precomputed after VAE adaptation and reused during DiT fine-tuning, avoiding repeated execution of the VAE encoder during transformer optimization.

The same one-node, eight-XPU allocation is used for the controlled VAE and DiT training experiments, providing a consistent computational environment across the principal ablations.

Chapter 6

Results and Discussion

This chapter evaluates the adaptation of the pretrained LTX-Video architecture to four-channel compressible-flow trajectories and examines how the resulting VAE representations affect downstream generative rollout. The experiments are organized around two related questions. First, we investigate how much of the pretrained VAE must be adapted to obtain accurate reconstruction of the physical fields and how this adaptation is affected by the reconstruction objective, learning rate, and spatial resolution. Second, we examine whether reconstruction accuracy alone is sufficient to characterize a VAE representation for subsequent DiT fine-tuning. To this end, KL regularization and DiT-guided self-distillation are evaluated both at the VAE reconstruction level and in the complete VAE–DiT pipeline.

The results show a clear distinction between these two objectives. Full-structure VAE fine-tuning substantially improves reconstruction, with the newly introduced pressure channel being particularly sensitive to the degree of encoder adaptation. However, differences in VAE reconstruction accuracy do not translate directly into differences in rollout accuracy. Moderate latent-space regularization can improve the downstream DiT even when the corresponding VAE reconstruction error is unchanged or slightly increased. The complete pipeline therefore provides an additional criterion for evaluating the suitability of an adapted latent representation beyond reconstruction quality alone.

The chapter first defines the common evaluation protocol and reference baseline in Section 6.1. The VAE adaptation strategies and training ablations are then evaluated in Sections 6.2 and 6.3. Section 6.4 examines latent-space regularization through KL regularization and DiT-guided self-distillation. The corresponding representations are then evaluated in the complete VAE–DiT rollout pipeline in Section 6.5, including quantitative and qualitative analysis. Finally, Section 6.6 discusses the implications of these results and the limitations of the study.

6.1 Evaluation Protocol and Baseline

Unless stated otherwise, the quantitative results in this chapter follow the held-out test protocol defined in Chapter 5. Results are reported on the 500-simulation test set using channel-mean variance-normalized root mean squared error (VRMSE) as the primary metric. All headline comparisons use 256×256 trajectories, except for the dedicated resolution ablation, where the VAE trained at 128×128 is evaluated directly on the same 256×256 test trajectories. Complete metric definitions, data partitions, and training configurations are provided in Chapter 5, while additional uncertainty information and complete numerical results are reported in Appendices B and C.

The reference VAE is the Full FT configuration defined in Chapter 5: full encoder–decoder fine-tuning using the $\text{RMSE}+H^1$ reconstruction objective. On the test set, it achieves a channel-mean VRMSE of

$$\text{VRMSE}_{\text{ch}} = 0.0800 \pm 0.0014, \tag{6.1}$$

with per-channel VRMSE values of 0.0727, 0.0707, 0.0704, and 0.1064 for ρ , m_x , m_y , and p , respectively. Pressure therefore remains the most difficult field to reconstruct even for the fully adapted VAE.

The full-pipeline experiments use the same evaluation principle but replace direct VAE reconstruction by conditional latent-trajectory generation through the fine-tuned DiT. Each VAE configuration considered in the pipeline study is paired with a separately fine-tuned DiT trained on that VAE's posterior-mean latent representation. All DiTs use identical training settings: 8,000 optimization steps, a learning rate of 10^{-5} , and an effective batch size of 32. At inference time, the initial physical frame is encoded using the posterior mean, matching the representation used during DiT training. The DiT then generates the remaining latent trajectory, which is decoded by the corresponding VAE. The baseline Full FT VAE combined with its fine-tuned DiT obtains a test-set channel-mean rollout VRMSE of 0.4984 ± 0.0060 . Two complementary quantities are therefore distinguished throughout the remainder of this chapter. *VAE-only* performance denotes encode–decode reconstruction of the complete ground-truth trajectory and represents the reconstruction floor of a particular VAE. *VAE+DiT* performance denotes generation from the initial frame through the complete latent rollout and decoding pipeline. Comparisons between different VAE latent spaces are based on the decoded physical-field errors rather than directly on latent-space VRMSE, since the scale and variance of the latent representation differ between independently adapted VAEs.

Finally, small differences between VAE configurations are interpreted in light of the observed reconstruction-training variability. A three-seed experiment at 128×128 indicated a variation of approximately 1.1%, while repeated training of the same 256×256 configuration on two compute systems differed by approximately 0.8%. Differences of roughly 1% or less in VAE reconstruction are therefore treated conservatively rather than as strong evidence of improved reconstruction performance.

6.2 VAE Adaptation to Four-Channel Physical Fields

The first set of experiments investigates how much of the pretrained LTX-Video VAE must be adapted to represent the four physical variables ρ , m_x , m_y , and p . This is not a direct application of the original RGB VAE: the input and output convolutions are widened from three to four channels, and the newly introduced fourth channel is associated with pressure. Its weights are zero-initialized, whereas the weights corresponding to the first three channels are inherited from the pretrained model. The experiments therefore progressively increase the amount of trainable VAE structure, from an unchanged pretrained representation to full encoder–decoder fine-tuning.

Figure 6.1 illustrates the four physical fields for an example Euler multi-quadrant trajectory. The evolution from the initial condition to later time steps contains sharp discontinuities and increasingly complex spatial structures, providing a qualitative view of the information that must be preserved by the adapted VAE.

Table C.1 summarizes the reconstruction performance of the adaptation variants on the 500-simulation test set. A clear trend is observed as progressively more of the pretrained VAE is allowed to adapt. The channel-mean VRMSE decreases from 0.5569 for the unfinetuned model to 0.4643 with the color adapter, 0.1441 with decoder-only fine-tuning, 0.0929 when the encoder input convolution is additionally trained, and finally 0.0800 with full-structure fine-tuning. The corresponding per-channel results show that this progression is driven particularly by the treatment of the newly introduced pressure channel.

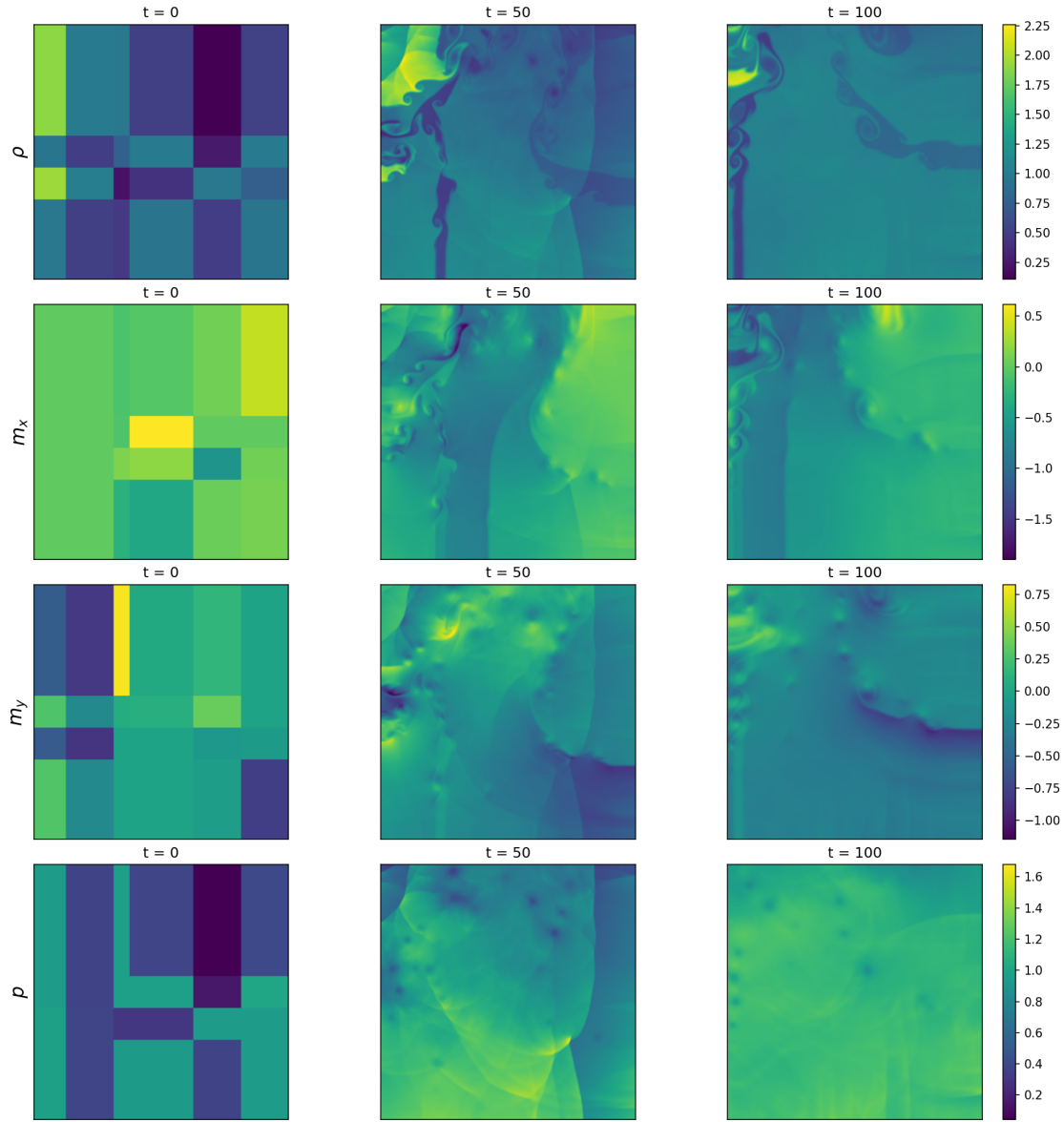


Figure 6.1: Example Euler multi-quadrant trajectory with $\gamma = 1.40$. Density ρ , horizontal momentum m_x , vertical momentum m_y , and pressure p are shown at three representative time steps ($t = 0, 50$, and 100), illustrating the evolution of the four-channel physical fields used for VAE adaptation.

6.2.1 Unadapted and Color-Adapter Baselines

The unfinetuned model provides a lower-complexity reference for evaluating the need for VAE adaptation. In this configuration, the pretrained three-channel input and output convolutions are expanded to four channels, but the VAE parameters are otherwise left unchanged. This model obtains a channel-mean VRMSE of 0.5569 ± 0.0117 , substantially above the Full FT reference value of 0.0800 ± 0.0014 .

The per-channel errors reveal that this poor overall result is not caused equally by all four physical variables. The VRMSE values for ρ , m_x , and m_y are 0.1269, 0.1116, and 0.1302, respectively, whereas the pressure error reaches 1.8589. This asymmetry follows from the four-channel initialization described in Chapter 4: the first three channel positions retain pretrained weights, while the newly introduced fourth channel is zero-initialized. Consequently, without adaptation, the pretrained VAE does not possess a learned pathway for accurately encoding and reconstructing the pressure field.

The color-adapter strategy tests whether the original RGB representation can instead be preserved while learning only lightweight mappings between the four physical channels and the pretrained three-channel interface. This reduces the channel-mean VRMSE from 0.5569 to 0.4643. The errors of the first three variables decrease to 0.1050, 0.0863, and 0.0924 for ρ , m_x , and m_y , respectively. Pressure, however, remains poorly reconstructed, with a VRMSE of 1.5736.

The adapter therefore improves the representation of the physical fields without modifying the pretrained VAE itself, but it does not resolve the dominant error associated with the additional channel. In particular, a learned $4 \leftrightarrow 3$ channel mapping is insufficient to obtain reconstruction quality comparable to a VAE whose internal representation is allowed to adapt to the four-field data.

6.2.2 Decoder-Side Adaptation

The next experiments increase the adaptation depth by fine-tuning the decoder. When the decoder is trained while the encoder remains frozen, the channel-mean VRMSE decreases substantially to 0.1441 ± 0.0023 . The largest improvement again occurs for pressure, whose VRMSE decreases from 1.5736 with the color adapter to 0.2980. The other three channels reach VRMSE values of 0.0974, 0.0870, and 0.0938 for ρ , m_x , and m_y , respectively.

Although decoder fine-tuning therefore recovers a large fraction of the reconstruction quality, the result remains 80.0% above the Full FT baseline in channel-mean VRMSE. This remaining gap is consistent with a limitation on the encoder side: adapting the decoder can improve the mapping from latent representations back to four physical fields, but the frozen encoder still cannot directly learn how the newly introduced pressure input should be represented in latent space.

This interpretation is supported by the intermediate *Decoder + conv_in* experiment. In addition to the decoder, this configuration makes the widened input convolution of the encoder trainable while leaving the remaining encoder structure frozen. The channel-mean VRMSE decreases further from 0.1441 to 0.0929 ± 0.0015 . Most notably, the pressure error decreases from 0.2980 to 0.1299. The errors of ρ , m_x , and m_y also decrease to 0.0843, 0.0772, and 0.0803, respectively.

Thus, allowing the encoder to learn even at its input interface closes most of the gap between decoder-only adaptation and full fine-tuning. Nevertheless, the resulting channel-mean error remains 16.1% above the Full FT baseline, indicating that adaptation beyond the first encoder layer provides an additional reconstruction benefit.

6.2.3 Full-Structure Fine-Tuning

Finally, all encoder and decoder parameters are jointly fine-tuned on the four-channel physical trajectories. This configuration achieves the lowest reconstruction error among the adaptation-depth experiments, with a channel-mean VRMSE of 0.0800 ± 0.0014 . The corresponding errors are 0.0727 for

density, 0.0707 for horizontal momentum, 0.0704 for vertical momentum, and 0.1064 for pressure. The progression across the adaptation experiments is therefore monotonic:

$$\begin{aligned} 0.5569 \text{ (unfintuned)} &> 0.4643 \text{ (color adapter)} > 0.1441 \text{ (decoder only)} \\ &> 0.0929 \text{ (decoder + conv_in)} > 0.0800 \text{ (full fine-tuning)}. \end{aligned} \quad (6.2)$$

Each increase in adaptation depth improves the channel-mean reconstruction error, and every intermediate adaptation strategy remains clearly above the Full FT result. The largest differences occur when the model is given the ability to learn how the fourth physical channel enters and leaves the latent representation.

Pressure nevertheless remains the most difficult field even after full fine-tuning, with a VRMSE of 0.1064 compared with approximately 0.07 for the other three channels. The substantial reduction from 1.8589 in the unfintuned model to 0.1064 after full fine-tuning shows that the additional physical channel cannot be accommodated adequately by interface expansion alone. At the same time, the remaining improvement from Decoder + conv_in to Full FT shows that adapting the deeper encoder representation provides a measurable benefit beyond learning only a new four-channel input projection.

Taken together, these experiments establish full-structure fine-tuning as the VAE adaptation strategy used for the subsequent training ablations. They also show that successful transfer from an RGB video VAE to the four-channel Euler representation requires adaptation on both sides of the latent bottleneck. The following sections therefore retain the fully trainable encoder–decoder architecture while varying the reconstruction objective, optimization settings, spatial resolution, and latent-space regularization.

6.3 VAE Training Ablations

Having established full-structure fine-tuning as the VAE adaptation strategy, the next experiments examine three training choices that affect reconstruction quality: the reconstruction objective, learning rate, and spatial training resolution. Each ablation modifies one component of the Full FT reference configuration introduced in Section 6.1, while retaining the same encoder–decoder adaptation strategy. The results are evaluated using the same channel-mean VRMSE on the held-out test set.

Overall, the ablations show that the VAE is relatively insensitive to a small modification of the reconstruction objective, but substantially more sensitive to the optimization rate and spatial resolution. The combined RMSE and H^1 objective provides a small improvement over RMSE alone, whereas adding SSIM degrades reconstruction. Reducing the learning rate leads to underperformance within the fixed training budget, while increasing it beyond the reference setting results in unstable training. Training at lower spatial resolution likewise produces a clear loss of reconstruction accuracy when evaluated at the target resolution.

6.3.1 Reconstruction Loss

The reference VAE is optimized using the combination of RMSE and the H^1 gradient loss defined in Chapter 4. To assess the contribution of the gradient term, an otherwise identical model is trained using RMSE alone. The RMSE-only model reaches a channel-mean VRMSE of 0.0814 ± 0.0014 , compared with 0.0800 ± 0.0014 for the RMSE+ H^1 reference configuration. This corresponds to a relative increase of 1.7% in reconstruction error.

Although the magnitude of this difference is close to the observed training-variability scale discussed in Section 6.1, the direction of the difference is highly consistent across the test set: the RMSE+ H^1 configuration achieves a lower channel-mean error on 491 of the 500 test simulations. The per-channel results show VRMSE values of 0.0767, 0.0718, 0.0724, and 0.1048 for ρ , m_x , m_y , and p with RMSE

alone, compared with 0.0727, 0.0707, 0.0704, and 0.1064 for the reference model. The H^1 term therefore provides a small and consistent overall benefit, rather than a large reduction in reconstruction error.

A second ablation augments the $\text{RMSE}+H^1$ objective with an SSIM term of weight 0.15. In contrast to the H^1 contribution, adding SSIM increases the channel-mean VRMSE to 0.0889 ± 0.0015 , which is 11.1% above the reference configuration. The degradation occurs across all four channels, with VRMSE values of 0.0808, 0.0787, 0.0786, and 0.1174 for ρ , m_x , m_y , and p , respectively.

These results do not indicate an advantage from adding SSIM to the physical-field reconstruction objective under the evaluated weighting. Consequently, the combination of RMSE and H^1 is retained as the reference reconstruction loss. The result should be interpreted as a comparison of the evaluated objectives and weights rather than as a general statement that SSIM-based objectives are unsuitable for physical fields.

6.3.2 Learning Rate

The learning-rate ablation evaluates the optimization sensitivity of full-structure VAE fine-tuning. The reference configuration uses a peak learning rate of 5×10^{-5} following 200 warm-up steps, followed by cosine decay to 10^{-6} . Reducing the peak learning rate to 10^{-5} while keeping the training duration fixed at 20 epochs increases the channel-mean VRMSE from 0.0800 ± 0.0014 to 0.0938 ± 0.0015 , corresponding to a relative degradation of 17.2%.

The effect is visible in every physical variable. At a peak learning rate of 10^{-5} , the per-channel VRMSE values are 0.0805, 0.0773, 0.0778, and 0.1395 for ρ , m_x , m_y , and p , respectively. Pressure is affected most strongly, increasing from 0.1064 in the reference configuration to 0.1395. Thus, within the common 20-epoch training budget, the lower learning rate does not reach the reconstruction quality obtained with 5×10^{-5} .

A higher peak learning rate of 10^{-4} was also tested but did not produce a usable model. Training became unstable shortly after the warm-up phase reached the peak learning rate: the validation VRMSE increased from approximately 0.15 to approximately 1.07, after which the model did not recover. Since this run failed to converge to a meaningful reconstruction solution, it is excluded from the quantitative comparison table and is treated as an optimization failure rather than as a competitive configuration.

Taken together, the learning-rate experiments place the reference value of 5×10^{-5} between a lower rate that converges more slowly within the available training budget and a higher rate that becomes unstable. It is therefore retained for the remaining VAE experiments.

6.3.3 Spatial Resolution

The spatial-resolution experiment examines whether the VAE can be adapted at reduced resolution without substantially compromising reconstruction at the target resolution. An otherwise identical Full FT model is trained on the 128×128 version of the Euler multi-quadrant dataset rather than on the 256×256 trajectories used by the reference configuration.

Because the VAE is fully convolutional, the model trained at 128×128 can be applied directly to 256×256 inputs. For the test comparison, it is therefore evaluated on exactly the same 500 256×256 simulations as the reference model, without spatial resampling during evaluation. The model retains the normalization statistics computed from the 128×128 training data; these differ from the corresponding 256×256 statistics by less than 0.2%.

Under this common test protocol, training at 128×128 results in a channel-mean VRMSE of 0.1200 ± 0.0019 , compared with 0.0800 ± 0.0014 for training at 256×256 . The lower-resolution training configuration therefore increases the reconstruction error by 49.9%. Its per-channel VRMSE values are

0.1202, 0.1009, 0.1016, and 0.1572 for ρ , m_x , m_y , and p , respectively, all above the corresponding values of the reference model.

The same conclusion is obtained when each model is evaluated at its training resolution during validation: the lower-resolution configuration remains approximately 50% worse. The degradation observed on the 256×256 test set is therefore not explained simply by applying the fully convolutional VAE at a resolution different from that used during training.

These results indicate that, for the present dataset and training setup, the computational savings obtained by training the VAE at 128×128 come with a substantial loss in reconstruction quality. The 256×256 resolution is consequently retained for the subsequent latent-regularization and VAE–DiT experiments. Together with the adaptation results of Section 6.2, the ablation study establishes the reference VAE configuration used in the remainder of this chapter: full encoder–decoder fine-tuning at 256×256 , using RMSE+ H^1 and a peak learning rate of 5×10^{-5} .

6.4 Latent-Space Regularization

The preceding experiments optimize the VAE primarily for reconstruction quality. However, in the complete latent generative pipeline, the VAE serves not only as a compressor and reconstructor but also defines the latent representation on which the DiT is trained. A representation that minimizes reconstruction error is therefore not necessarily the most suitable representation for downstream generative modeling.

To investigate this distinction, two forms of latent-space regularization are applied to the Full FT VAE established in Sections 6.2 and 6.3. The first introduces conventional KL regularization of the VAE posterior. The second uses DiT-guided self-distillation (SDS), in which the stock pretrained LTX-Video transformer provides a latent-space training signal while remaining frozen. In both cases, the reconstruction architecture, data, and optimization settings are otherwise kept consistent with the Full FT reference configuration.

This section first evaluates how these regularizers affect VAE reconstruction and the learned latent representation. Their effect on the complete VAE–DiT rollout is deliberately considered separately in Section 6.5, where a new DiT is fine-tuned for each regularized VAE. This separation is important because reconstruction quality and downstream generative performance need not follow the same ordering.

6.4.1 KL Regularization

The Full FT reference model is optimized without an explicit KL penalty. Under this reconstruction-dominated objective, the posterior variance becomes very small, producing an effectively near-deterministic latent representation. KL regularization is introduced to constrain the posterior distribution while preserving as much reconstruction accuracy as possible.

Two successfully trained KL weights are evaluated, $\lambda_{\text{KL}} = 10^{-7}$ and 10^{-5} . With the weaker regularization, the channel-mean test VRMSE is 0.0804 ± 0.0014 , compared with 0.0800 ± 0.0014 for the unregularized Full FT reference. The relative increase is only 0.5%, which lies within the observed VAE training-variability scale. The per-channel VRMSE values are 0.0728, 0.0707, 0.0705, and 0.1077 for ρ , m_x , m_y , and p , respectively, and are correspondingly close to those of the reference model.

Despite the nearly unchanged reconstruction performance, $\lambda_{\text{KL}} = 10^{-7}$ has a substantial effect on the posterior statistics. On the validation set, the KL term decreases from approximately 7.7×10^5 for the unregularized model to approximately 8.14×10^2 . Thus, a weak KL penalty substantially changes the posterior behavior while leaving physical-field reconstruction essentially unchanged.

Increasing the KL weight to 10^{-5} produces a stronger trade-off. The channel-mean test VRMSE rises to 0.0877 ± 0.0014 , corresponding to a 9.6% increase relative to the reference model. The per-channel

errors increase to 0.0807, 0.0792, 0.0784, and 0.1126 for ρ , m_x , m_y , and p , respectively. At the same time, the validation KL term decreases further to approximately 55.

These results demonstrate that posterior regularization can be introduced at very different reconstruction costs depending on its strength. A weak KL penalty strongly alters the posterior statistics without measurably changing reconstruction quality at the scale resolved by the present experiments, whereas the stronger penalty sacrifices part of the reconstruction accuracy in exchange for substantially stronger posterior regularization.

An intermediate experiment with $\lambda_{\text{KL}} = 10^{-6}$ became unstable shortly after the learning-rate warm-up and did not recover. Since the neighboring 10^{-7} and 10^{-5} configurations both trained normally, this run is treated as an optimization failure rather than evidence of a non-monotonic dependence on the KL weight, and it is excluded from the quantitative comparison.

6.4.2 DiT-Guided Self-Distillation

The second regularization strategy introduces a training signal from the pretrained LTX-Video transformer. In contrast to KL regularization, which directly constrains the posterior distribution, DiT-guided self-distillation encourages the adapted VAE representation to remain compatible with latent structure recognized by the pretrained video generative model. The transformer used to provide this signal is the stock, frozen LTX-Video model and is not fine-tuned on the Euler multi-quadrant dataset during VAE adaptation. The formulation of the SDS objective is described in Chapter 4.

Three SDS weights are evaluated: $\lambda_{\text{SDS}} = 10^{-4}$, 10^{-2} , and 1. At the two lower weights, reconstruction remains essentially unchanged relative to the Full FT reference. For $\lambda_{\text{SDS}} = 10^{-4}$, the channel-mean VRMSE is 0.0793 ± 0.0014 , a relative difference of -0.9% . For $\lambda_{\text{SDS}} = 10^{-2}$, the corresponding value is 0.0806 ± 0.0014 , or $+0.8\%$ relative to the reference. Both differences are within the approximately 1% training-variability scale identified in Section 6.1 and should therefore be interpreted as reconstruction ties rather than meaningful improvements or degradations.

The per-channel results lead to the same conclusion. At $\lambda_{\text{SDS}} = 10^{-4}$, the VRMSE values are 0.0723, 0.0703, 0.0700, and 0.1046 for ρ , m_x , m_y , and p , respectively. At $\lambda_{\text{SDS}} = 10^{-2}$, they are 0.0734, 0.0714, 0.0709, and 0.1069. Neither configuration therefore produces a substantial change in physical-field reconstruction relative to the unregularized Full FT VAE.

The behavior changes at the substantially stronger weight $\lambda_{\text{SDS}} = 1$. The channel-mean VRMSE increases to 0.1040 ± 0.0015 , which is 29.9% above the Full FT reference. The corresponding per-channel errors increase to 0.0963, 0.0941, 0.0940, and 0.1316. Strong SDS regularization therefore competes with the reconstruction objective sufficiently to degrade the VAE’s ability to preserve the physical fields.

Taken at the VAE level alone, these experiments identify a range in which latent regularization can modify the training objective without materially sacrificing reconstruction. Weak KL regularization and moderate SDS regularization retain reconstruction performance close to the Full FT reference, whereas stronger regularization introduces an increasingly visible reconstruction cost. Importantly, reconstruction error alone does not establish whether these regularized representations are more or less suitable for generative rollout. That question requires fine-tuning and evaluating the DiT on each corresponding latent space and is addressed in Section 6.5.

6.5 Full VAE–DiT Pipeline

The preceding sections evaluate how different adaptation and regularization strategies affect VAE reconstruction. The ultimate surrogate task, however, requires generation of the future physical trajectory

rather than reconstruction of an already observed sequence. This section therefore evaluates the complete VAE–DiT pipeline.

For each VAE considered in the pipeline study, a separate DiT is fine-tuned on the posterior-mean latents produced by that VAE. The DiT training configuration is kept identical across the experiments, as described in Section 6.1. During inference, only the initial physical frame is provided. It is encoded using the posterior mean, the DiT generates the remaining latent trajectory, and the corresponding VAE decoder maps the generated latents back to the four physical fields.

The experiments compare the unfinetuned VAE, the Full FT reference VAE, two KL-regularized VAEs, and three SDS-regularized VAEs. This comparison makes it possible to distinguish two sources of error: the reconstruction floor imposed by the VAE and the additional error introduced when the future trajectory must be generated in latent space. It also provides a direct test of whether the VAE configuration with the lowest reconstruction error necessarily provides the most suitable latent representation for downstream generative modeling.

6.5.1 Physical-Field Rollout Performance

Table C.3 reports the complete pipeline results on the 500 held-out test simulations. The Full FT reference VAE has a VAE-only channel-mean VRMSE of 0.0802 under the pipeline evaluation protocol. After replacing the ground-truth latent trajectory with a DiT-generated trajectory, the error increases to 0.4984 ± 0.0060 . The rollout error is therefore approximately 6.2 times the corresponding VAE reconstruction floor.

This gap is observed for all fine-tuned VAE configurations. The VAE+DiT channel-mean VRMSE ranges from 0.4583 to 0.5678, whereas the corresponding VAE-only errors range from approximately 0.08 to 0.10. The remaining error of the complete surrogate is therefore dominated by the generative stage rather than by direct VAE reconstruction. Improving VAE reconstruction by a few percent cannot by itself eliminate the much larger error introduced during latent rollout.

Fine-tuning the VAE nevertheless remains essential. When the unfinetuned VAE is used, its VAE-only error is already 0.5579, and the complete pipeline reaches 0.9501 ± 0.0128 , which is 90.6% above the Full FT pipeline. The pressure channel is particularly poor, with a rollout VRMSE of 1.859. This behavior is consistent with the four-channel adaptation results of Section 6.2: without VAE fine-tuning, the newly introduced pressure channel is not represented adequately. The comparatively small ratio of 1.7 between rollout error and reconstruction floor for this model should therefore not be interpreted as favorable DiT behavior; it results primarily from the already poor reconstruction floor.

The latent-regularized VAEs produce a different pattern. With $\lambda_{\text{KL}} = 10^{-7}$, the pipeline reaches 0.4891 ± 0.0059 , a 1.9% reduction relative to the Full FT reference. Increasing the KL weight to 10^{-5} reduces the rollout error further to 0.4733 ± 0.0053 , corresponding to a 5.0% improvement.

For SDS, the weakest weight, $\lambda_{\text{SDS}} = 10^{-4}$, produces 0.4974 ± 0.0060 , essentially unchanged from the reference pipeline. The intermediate weight, $\lambda_{\text{SDS}} = 10^{-2}$, obtains the lowest rollout error among the evaluated configurations, 0.4583 ± 0.0054 , corresponding to an 8.1% reduction relative to the Full FT reference. In contrast, strong SDS regularization with $\lambda_{\text{SDS}} = 1$ increases the rollout error to 0.5678 ± 0.0051 , or 13.9% above the reference.

The per-channel errors show that the distribution of error also changes between VAE reconstruction and full rollout. For the Full FT pipeline, the VRMSE values are 0.514, 0.506, 0.516, and 0.459 for ρ , m_x , m_y , and p , respectively. For the $\lambda_{\text{SDS}} = 10^{-2}$ configuration, these values decrease to 0.483, 0.465, 0.474, and 0.411. Unlike the VAE-only results, where pressure is consistently the most difficult channel, pressure has the lowest rollout error among the four fields for the fine-tuned pipelines. The dominant downstream error is therefore no longer the fourth-channel reconstruction problem identified during VAE adaptation, but is distributed across the generated density and momentum fields.

Paired comparisons over the 500 test simulations support the consistency of the larger pipeline differences. The $\lambda_{\text{SDS}} = 10^{-2}$ configuration improves upon the Full FT reference on 443 of the 500

simulations, while $\lambda_{\text{KL}} = 10^{-5}$ does so on 369. In contrast, $\lambda_{\text{SDS}} = 10^{-4}$ is better on 261 simulations and its paired Wilcoxon comparison with the reference is not significant ($p = 0.38$). The other paired comparisons yield $p < 10^{-14}$. These paired tests therefore provide additional evidence that the larger observed differences are distributed across the fixed test set rather than being driven by a small number of individual simulations.

6.5.2 Reconstruction Quality versus Generative Performance

A central observation of the pipeline experiments is that VAE reconstruction quality does not determine downstream generative performance. The ordering of the models by VAE-only error differs from their ordering after DiT fine-tuning and rollout.

This distinction is particularly clear for the KL-regularized models. As shown in Section 6.4.1, $\lambda_{\text{KL}} = 10^{-5}$ increases the VAE reconstruction error by 9.6% relative to Full FT. Under the pipeline evaluation protocol, its VAE-only error is 0.0879, compared with 0.0802 for Full FT. Nevertheless, after separately fine-tuning a DiT on each representation, the KL-regularized model obtains a rollout error of 0.4733, compared with 0.4984 for Full FT. A representation that is measurably worse for reconstruction therefore yields a 5.0% lower downstream rollout error.

The SDS experiments provide an even clearer separation between the two criteria. At $\lambda_{\text{SDS}} = 10^{-2}$, the VAE-only error is 0.0808, essentially tied with the Full FT value of 0.0802, whereas the rollout error decreases from 0.4984 to 0.4583. Thus, an approximately unchanged reconstruction floor is accompanied by an 8.1% reduction in the complete pipeline error.

Conversely, stronger regularization does not lead monotonically to better generation. With $\lambda_{\text{SDS}} = 1$, the VAE-only error rises to 0.1044 and the rollout error to 0.5678. The effect of SDS is therefore non-monotonic under the evaluated weights: a weak regularizer has essentially no downstream effect, an intermediate weight improves the pipeline while preserving reconstruction, and an excessively strong weight degrades both objectives.

These comparisons indicate that reconstruction accuracy is a necessary but insufficient criterion for selecting a VAE for the present latent generative surrogate. The VAE must reconstruct the physical fields accurately enough to provide a low error floor, but the structure of its latent representation also affects how successfully a DiT can learn the trajectory distribution. Moderate KL or SDS regularization can improve the latter property without improving reconstruction itself.

The latent-space VRMSE reported for diagnostic purposes provides a useful example of why decoded physical-field performance is required for cross-VAE comparison. The Full FT and KL- 10^{-7} models both obtain a latent VRMSE of approximately 0.558, while KL- 10^{-5} decreases it to 0.273 and SDS- 10^{-2} to 0.290. Most strikingly, SDS-1 obtains the lowest latent VRMSE, 0.065, despite having the highest physical-field rollout error among the fine-tuned VAE configurations.

These latent VRMSE values must not be interpreted as a direct ranking of the different VAEs. Each VAE defines its own latent space, and the metric is normalized by the variance of that particular representation. Consequently, its scale is not common across independently adapted VAEs. The decoded VAE+DiT channel-mean VRMSE is therefore used as the primary measure of downstream performance. The results are consistent with the hypothesis that moderate latent regularization produces representations that are better conditioned for the generative model. KL regularization prevents the posterior from collapsing toward a near-deterministic representation, while SDS exposes the VAE during adaptation to a training signal derived from the frozen pretrained video transformer. However, the present experiments do not isolate the precise mechanism responsible for the downstream improvement. Latent compatibility is therefore treated as an interpretation supported by the observed pipeline behavior rather than as a directly established causal mechanism.

6.5.3 Dependence on the Physical Parameter γ

The test set contains 50 simulations for each of the ten discrete values of the adiabatic index γ , allowing the pipeline performance to be examined across the conditioning parameter. The per- γ results show that the overall improvements from latent regularization are not restricted to a small subset of the parameter values.

For the Full FT reference pipeline, the channel-mean VRMSE varies from 0.470 at $\gamma = 1.404$ to 0.540 at $\gamma = 1.597$. Relatively high errors are also observed at the lower and upper extremes of the evaluated range, including 0.526 at $\gamma = 1.130$ and 0.521 at $\gamma = 1.760$. This suggests that the more extreme parameter values are among the more difficult cases for the learned rollout, although the dependence is not strictly monotonic in γ .

The SDS- 10^{-2} configuration improves upon the Full FT reference at all ten evaluated values of γ . Its channel-mean VRMSE values are 0.495, 0.463, 0.443, 0.451, 0.450, 0.456, 0.435, 0.452, 0.483, and 0.456 as γ increases from 1.130 to 1.760. The improvement is therefore distributed across the complete evaluated parameter set rather than being produced by isolated test groups.

The gains are particularly pronounced for some of the more difficult parameter values. At $\gamma = 1.760$, for example, the Full FT reference has a VRMSE of 0.521, whereas SDS- 10^{-2} reduces it to 0.456, a relative reduction of approximately 12.5%. KL- 10^{-5} reaches 0.472 at the same parameter value, corresponding to a reduction of approximately 9.4%. At $\gamma = 1.597$, the corresponding values are 0.540, 0.483, and 0.486 for Full FT, SDS- 10^{-2} , and KL- 10^{-5} , respectively.

The VAE-only reconstruction floor varies much less strongly with γ . For the Full FT VAE it ranges from approximately 0.076 to 0.094 across the ten parameter groups, remaining far below the corresponding rollout errors. The dependence on γ observed in the complete pipeline is therefore dominated by trajectory generation rather than by direct reconstruction.

These results indicate that the benefit of moderate latent regularization persists across the conditioning range represented in the test set. In particular, the improvement of SDS- 10^{-2} at every evaluated γ provides evidence that its aggregate reduction in rollout error is not driven by a single parameter subset. The present experiment does not, however, evaluate interpolation to unseen values of γ , since the test set uses the same ten discrete parameter values represented in the dataset.

6.5.4 Qualitative Rollout Analysis

Figure 6.2 provides a qualitative comparison of VAE reconstruction and full VAE+DiT rollout for an example trajectory. The example corresponds to simulation 2507 with $\gamma = 1.40$ and shows the physical fields at frame 100. The columns compare the ground truth, direct VAE reconstruction, VAE+DiT rollout, and the corresponding reconstruction and rollout residuals.

The VAE-only reconstruction preserves the principal spatial structures of all four fields closely. The strongest residuals are concentrated near sharp flow features, where small spatial displacements can produce localized pointwise errors. Away from these structures, the reconstructed fields remain close to the ground truth. This visual behavior is consistent with the low VAE reconstruction error obtained in the quantitative evaluation.

The VAE+DiT prediction retains the overall organization of the flow but shows substantially larger deviations from the target state. In contrast to the localized VAE reconstruction residuals, the rollout errors extend over broader spatial regions and are visible in multiple physical fields. The difference between the two error patterns is consistent with the quantitative finding that the dominant error in the complete pipeline is introduced during latent trajectory generation rather than during decoding alone.

For this complete simulation, the channel-mean VRMSE is 0.0930 for VAE-only reconstruction and 0.4668 for the VAE+DiT rollout. These values characterize the displayed example and are consistent with the substantially larger rollout errors observed in the aggregate evaluation.

The qualitative comparison illustrates the main distinction established by the quantitative experiments.

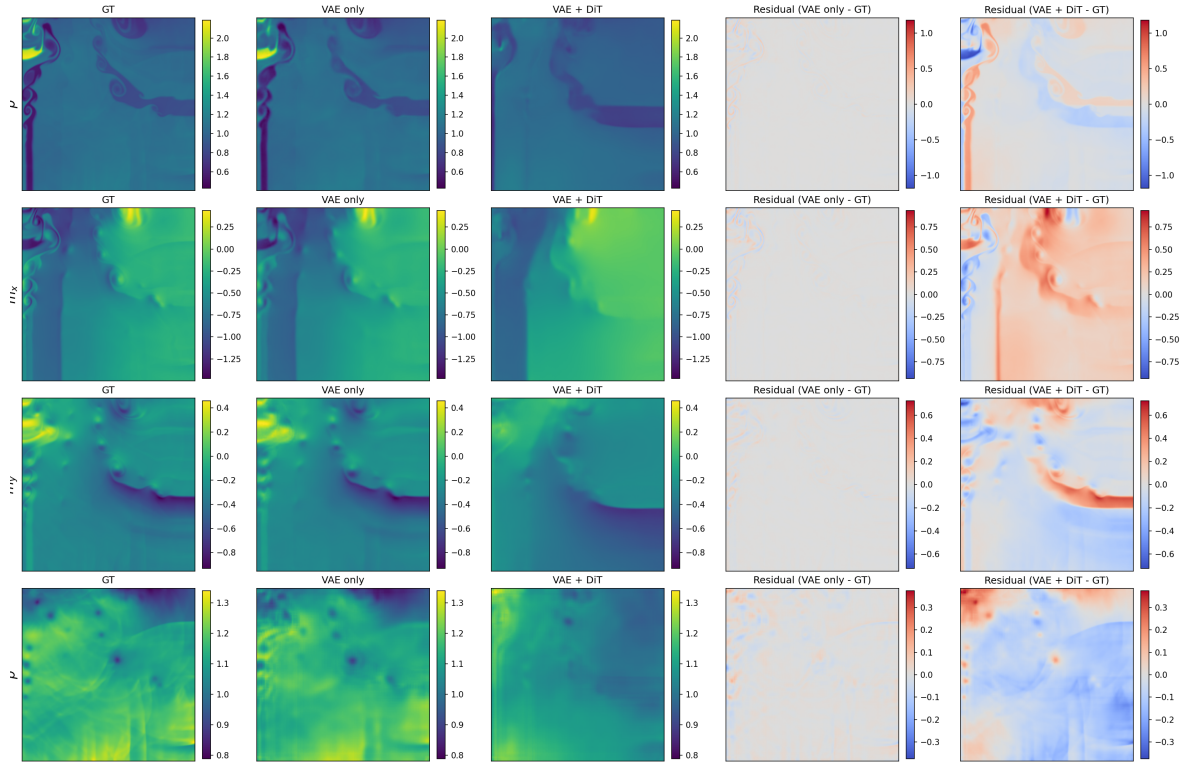


Figure 6.2: Qualitative comparison of VAE reconstruction and VAE-DiT rollout for simulation 2507 ($\gamma = 1.40$) at frame 100. From left to right, the columns show the ground truth, VAE-only reconstruction, VAE+DiT prediction, VAE reconstruction residual, and VAE+DiT rollout residual for the four physical fields. The VAE reconstruction closely follows the target, whereas the generated rollout exhibits substantially larger spatial deviations.

The adapted VAE can reconstruct complex flow structures with comparatively small error, whereas accurate long-horizon generation remains substantially more difficult. At the same time, the regularization experiments show that the quality of this generative stage depends not only on the reconstruction floor but also on the latent representation on which the DiT is trained.

6.6 Discussion

The experiments in this chapter demonstrate that a pretrained latent video model can be adapted to four-channel compressible-flow data, but they also show that successful adaptation involves two distinct requirements. The VAE must first provide an accurate representation of the physical fields, and the resulting latent space must then support effective learning by the generative model. These requirements are related but not equivalent.

The VAE experiments establish that accurate reconstruction requires substantial adaptation of the pretrained representation. Once this is achieved, however, further improvements in reconstruction are not sufficient to predict the behavior of the complete surrogate. The full-pipeline experiments instead reveal a separation between reconstruction quality and generative performance, with moderate latent regularization improving downstream rollout even when VAE reconstruction remains unchanged or becomes slightly worse. The following discussion considers the implications of these observations for transferring pretrained video models to physical simulation.

6.6.1 Adapting Pretrained Video Representations to Physical Fields

The VAE results provide evidence that a pretrained video representation can be repurposed for physical-field compression, despite substantial differences between natural RGB video and the Euler multi-quadrant data considered here. The target trajectories contain four continuous physical variables rather than three color channels, and their spatial structures are governed by compressible-flow dynamics rather than by the statistics of natural images. Nevertheless, after full-structure fine-tuning, the pretrained VAE reconstructs all four fields with a channel-mean VRMSE of approximately 0.08.

At the same time, the adaptation-depth experiments show that this transfer cannot be achieved by modifying only the external channel interface. The unfinetuned and color-adapter configurations remain substantially worse than the fully adapted VAE, with the newly introduced pressure channel accounting for a large fraction of the initial reconstruction error. Decoder-only fine-tuning produces a major improvement, and allowing the encoder input convolution to adapt closes much of the remaining gap. Full encoder–decoder fine-tuning nevertheless provides the lowest reconstruction error among the evaluated adaptation strategies.

This progression suggests that the pretrained model provides a useful initialization, but that the internal representation must be allowed to reorganize for the statistics of the physical fields. In particular, the success of full fine-tuning should not be interpreted as evidence that the original RGB channels already correspond to specific physical variables. The first three channel positions inherit pretrained weights whereas the fourth pressure channel is newly introduced and zero-initialized. The adaptation experiment therefore does not isolate semantic transfer from RGB to individual Euler variables.

The resolution experiment further indicates that the quality of this adaptation depends on retaining sufficient spatial information during training. Training at 128×128 and evaluating on the common 256×256 test set produces substantially higher reconstruction error than training directly at 256×256 . This is particularly relevant for compressible flow, where shocks and other sharp spatial structures must be represented accurately. Although the fully convolutional VAE can technically operate across these resolutions, architectural applicability across resolutions does not imply equivalent reconstruction quality.

Taken together, these results support the feasibility of adapting a large pretrained video VAE to the

present four-channel physical domain. However, the useful transferable component appears to be better understood as a pretrained initialization that can be extensively adapted, rather than as a fixed natural-video representation that can be reused without substantial modification.

6.6.2 Reconstruction Accuracy and Latent Compatibility

The most important distinction emerging from the full-pipeline experiments is between reconstruction accuracy and downstream generative performance. For the fine-tuned VAEs, direct reconstruction reaches channel-mean VRMSE values close to 0.08–0.10, whereas the complete VAE–DiT pipeline produces errors of approximately 0.46–0.57. The dominant remaining error is therefore associated with generation of the future latent trajectory rather than with direct encode–decode reconstruction.

More importantly, the ordering of the VAE configurations by reconstruction error does not match their ordering in the complete pipeline. The unregularized Full FT VAE provides an accurate reconstruction baseline, but it does not produce the lowest observed rollout error. KL regularization with $\lambda_{\text{KL}} = 10^{-5}$ worsens VAE reconstruction while reducing downstream rollout error, whereas SDS with $\lambda_{\text{SDS}} = 10^{-2}$ leaves reconstruction approximately unchanged while producing the lowest rollout error among the evaluated configurations.

This observation has an important methodological consequence. Evaluating a VAE only by reconstruction error is insufficient when the VAE is intended to define the representation space of a latent generative model. A reconstruction objective primarily requires that the encoder and decoder form an accurate pair. It does not by itself require the distribution or geometry of the resulting latent representations to be easy for a separate generative model to learn.

The latent-VRMSE results reinforce this distinction. Strong SDS regularization produces the lowest normalized latent error among the evaluated configurations while simultaneously producing worse decoded physical-field rollout. Since independently adapted VAEs define different latent distributions and the reported latent error is normalized using each representation’s own variance, these latent errors do not provide a common cross-VAE scale. Decoded physical-field error is therefore the more meaningful criterion for comparing complete surrogate pipelines.

In this thesis, *latent compatibility* is used to describe this downstream requirement: the latent representation should not only permit accurate reconstruction but should also support effective trajectory generation by the DiT. The experiments demonstrate that these two properties can vary independently. They do not, however, directly identify a single geometric or statistical quantity that completely characterizes compatibility. The term should therefore be understood as an operational property measured through downstream VAE–DiT performance rather than as a fully identified latent-space mechanism.

6.6.3 Role of Latent Regularization

The KL and SDS experiments provide further evidence for the distinction between reconstruction quality and downstream suitability. Both methods modify the learned latent representation, but they do so through different training signals.

Without explicit KL regularization, reconstruction-oriented VAE fine-tuning drives the posterior toward very small variance. A weak KL penalty changes these posterior statistics substantially while leaving reconstruction essentially unchanged. A stronger KL weight introduces a measurable reconstruction cost, but the corresponding representation produces lower rollout error after DiT fine-tuning. This result suggests that minimizing reconstruction error alone can favor posterior properties that are not necessarily optimal for subsequent generative modeling.

SDS provides a complementary form of regularization. Instead of explicitly constraining the posterior toward a prescribed distribution, it introduces a training signal from the frozen pretrained LTX-Video transformer. Moderate SDS regularization can therefore influence the VAE representation without requiring

a large change in physical-field reconstruction. In the evaluated experiments, $\lambda_{\text{SDS}} = 10^{-2}$ preserves approximately the same reconstruction quality as the unregularized Full FT model while producing the lowest observed full-pipeline VRMSE.

The response to SDS is clearly non-monotonic. A weight of 10^{-4} has little measurable downstream effect, 10^{-2} improves the pipeline, and a weight of 1 degrades both reconstruction and rollout. The objective must therefore balance the physical reconstruction task against the additional latent-space constraint; increasing the regularization strength indiscriminately does not improve compatibility.

One possible interpretation of these results is that moderate regularization produces a better-conditioned latent representation for flow matching. KL regularization prevents the posterior from becoming almost deterministic and strongly changes its scale statistics, while SDS encourages the adapted VAE to retain structure that the pretrained video transformer can process. Under this interpretation, the downstream DiT benefits from a representation that is somewhat more constrained than the one obtained from reconstruction optimization alone.

The experiments, however, do not establish this mechanism directly. The improved rollout could depend on several interacting properties of the latent distribution, and the present study does not isolate which of them is responsible. In particular, no single latent-space diagnostic evaluated here predicts the downstream ordering of all models. The strongest conclusion is therefore empirical: moderate latent regularization can improve downstream DiT performance without improving VAE reconstruction, demonstrating that the VAE should be evaluated as part of the complete generative pipeline.

6.6.4 Limitations and Negative Results

Several limitations constrain the interpretation and generality of the results.

First, all DiTs are trained for the same 8,000 optimization steps using identical hyperparameters. This controlled protocol is useful for comparing the effect of changing the underlying VAE representation, but it does not guarantee that each DiT has converged or that the same optimization settings are optimal for every latent space. The reported results should therefore be interpreted as comparisons under a common training budget rather than as the best achievable performance of each configuration.

Second, the study is restricted to the Euler multi-quadrant dataset and to a single scalar conditioning parameter, γ . The test set contains unseen simulations but uses the same ten discrete γ values represented in the dataset. The per- γ results therefore demonstrate consistency across the evaluated parameter groups, but they do not establish interpolation or extrapolation to unseen values of γ , nor do they establish generalization to other PDE systems or flow regimes.

Third, the present work evaluates surrogate accuracy but does not provide a wall-clock comparison against the numerical solver used to generate the Euler trajectories. The experiments therefore establish reconstruction and rollout behavior, but not an end-to-end computational speedup relative to the underlying CFD solver.

Several negative results also inform the final experimental design. A peak VAE learning rate of 10^{-4} and the KL- 10^{-6} run became unstable and were excluded from the quantitative comparisons rather than interpreted as valid performance points. Adding SSIM with the evaluated weight degraded reconstruction relative to RMSE+ H^1 . Strong SDS regularization likewise degraded both VAE reconstruction and downstream rollout. These results show that neither additional loss terms nor stronger regularization are automatically beneficial and emphasize the need to evaluate their effects under the complete training and inference protocol.

Despite these limitations, the experiments establish two main practical findings for the present setting. First, adapting the pretrained video VAE deeply enough to represent all four physical fields is necessary for accurate reconstruction. Second, once a sufficiently accurate VAE has been obtained, reconstruction error alone is not sufficient for selecting the latent representation used by the DiT. Evaluating VAE design choices through the complete downstream pipeline is therefore essential when adapting latent video generation architectures as surrogate models for physical dynamics.

Chapter 7

Conclusion and Future Work

7.1 Summary

This thesis investigated whether a pretrained latent video generation model can be adapted from natural video to surrogate modeling of two-dimensional compressible Euler flows. Starting from LTX-Video 2B, the original RGB representation was extended to four physical variables—density ρ , horizontal momentum m_x , vertical momentum m_y , and pressure p —while the heat-capacity ratio γ was introduced as an additional conditioning parameter.

The adaptation problem was studied at two coupled levels. First, the VAE was adapted to encode and reconstruct the four-channel physical trajectories. Second, the DiT was fine-tuned on the resulting latent representation to generate future trajectories conditioned on the initial state and γ . This two-stage formulation made it possible to distinguish reconstruction quality from the suitability of the learned latent representation for downstream generative modeling.

The experiments show that successful transfer requires substantial adaptation of the pretrained VAE and that accurate reconstruction alone is not sufficient for accurate physical rollout. In particular, changes to the latent representation can improve downstream DiT performance even when VAE reconstruction remains approximately unchanged or becomes moderately worse. The central conclusion of this thesis is therefore that a VAE used in a latent physical surrogate has two roles: it must represent the physical fields accurately, and it must define a latent space in which the generative model can effectively learn the physical dynamics.

7.2 Main Findings

The first main finding is that adaptation of the internal VAE representation is necessary for transferring LTX-Video from RGB video to four-channel Euler trajectories. The unfinetuned four-channel model obtains a channel-mean test VRMSE of 0.5569, while decoder-only fine-tuning reduces the error to 0.1441, adapting the encoder input convolution further reduces it to 0.0929, and full encoder–decoder fine-tuning achieves 0.0800. The pressure channel shows the strongest effect, with its VRMSE decreasing from 1.8589 to 0.1064. These results indicate that adapting only the channel interface is insufficient; the internal representation itself must adjust to the physical domain.

Second, the VAE experiments establish full encoder–decoder fine-tuning at 256×256 resolution with the $\text{RMSE}+H^1$ objective as the reference reconstruction configuration. The ablations further show that adaptation depth and spatial resolution have substantially larger effects on reconstruction than the relatively small difference between RMSE and $\text{RMSE}+H^1$. The evaluated SSIM-augmented objective does not improve reconstruction under the tested weighting.

Third, accurate VAE reconstruction does not imply accurate generative rollout. The Full FT VAE reconstructs the physical trajectories with a channel-mean VRMSE of approximately 0.08, whereas the corresponding VAE–DiT pipeline obtains a rollout VRMSE of 0.4984. The dominant remaining error of the surrogate is therefore associated with latent trajectory generation rather than direct VAE reconstruction.

Most importantly, reconstruction quality and downstream generative performance do not follow the same ordering across adapted VAE representations. With $\lambda_{\text{KL}} = 10^{-5}$, VAE reconstruction becomes moderately worse than the unregularized Full FT reference, while the rollout VRMSE decreases from 0.4984 to 0.4733. With $\lambda_{\text{SDS}} = 10^{-2}$, reconstruction remains approximately unchanged while the rollout VRMSE decreases further to 0.4583, an 8.1% reduction relative to the Full FT reference. The improvement is observed across all ten evaluated values of γ . Stronger regularization, however, does not continue this trend and can degrade both reconstruction and generation.

These results motivate a distinction between *reconstruction quality* and *generative suitability*. A latent representation that minimizes VAE reconstruction error is not necessarily the representation on which a separate generative model can most effectively learn the trajectory distribution. This conclusion is reinforced by the latent-space evaluation: normalized latent VRMSE does not provide a reliable common comparison across independently adapted VAEs, since each VAE defines a different latent distribution. Cross-VAE evaluation should therefore ultimately be performed in the decoded physical space.

Overall, the results show that the VAE should not be treated only as a compression module in a latent physical surrogate. Its representation also forms the learning space of the downstream generative model. VAE adaptation and selection should therefore be evaluated in the context of the complete VAE–DiT pipeline rather than solely through reconstruction metrics.

7.3 Limitations

The conclusions of this thesis are subject to several limitations. First, the experimental study is restricted to the Euler Multi-Quadrants dataset and therefore to a single family of compressible-flow problems. The same ten discrete values of the heat-capacity ratio γ occur in both training and testing. The experiments consequently demonstrate generalization to unseen trajectories within the evaluated parameter set, but do not establish interpolation or extrapolation to unseen values of γ or transfer to different governing equations and physical systems.

Second, the DiT comparisons use a common training budget of 8,000 optimization steps and identical training settings across the evaluated VAE representations. This controlled protocol isolates differences associated with the latent representation, but it does not establish the individually optimized performance limit of each VAE–DiT configuration. The absolute rollout errors should therefore be interpreted as results under a common fine-tuning protocol.

Third, although moderate KL regularization and DiT-guided self-distillation improve downstream rollout performance, the experiments do not establish the mechanism responsible for this improvement. They support the interpretation that properties of the adapted latent representation affect the difficulty of downstream trajectory generation, but do not identify a single measurable latent-space property that explains the observed behavior. Latent compatibility is therefore used operationally in this thesis to describe the suitability of a representation for downstream generation, as reflected by complete-pipeline performance. Finally, this work focuses primarily on predictive accuracy and representation adaptation. No systematic wall-clock comparison with the numerical solver used to generate the dataset was performed. The results therefore do not by themselves establish an end-to-end computational speed-up over conventional CFD simulation.

7.4 Future Work

The most immediate direction for future work is to improve latent trajectory generation. The remaining gap between VAE reconstruction and complete VAE–DiT rollout indicates that the DiT stage is the dominant source of error in the present surrogate. Longer or better optimized training and alternative flow-matching formulations could help determine how much of this gap can be reduced without changing the underlying VAE representation.

A second direction is to characterize latent compatibility more directly. The present results show that reconstruction accuracy and normalized latent prediction error are insufficient for predicting downstream generative performance across adapted VAEs. Future work could therefore investigate properties such as latent scale, variance and covariance structure, smoothness, and decoder sensitivity to latent perturbations, and relate these properties to flow-matching difficulty and decoded physical-space error. A useful latent-space diagnostic could reduce the need to fine-tune and evaluate a complete DiT for every candidate VAE representation.

A third direction is to test the transferability of the approach beyond the present experimental setting. Evaluation on unseen values of γ would provide a direct test of parameter interpolation and extrapolation, while experiments with different initial conditions, flow regimes, conservation laws, or PDE families would help determine whether the observed adaptation behavior generalizes beyond Euler Multi-Quadrants.

Finally, the practical value of the approach as a simulation surrogate should be evaluated against conventional numerical methods. Such an evaluation should consider physical prediction error together with inference time, memory requirements, training cost, and dataset-generation cost. This would clarify under which conditions adapting a pretrained generative video model provides a meaningful computational advantage for physical simulation.

In summary, adapting pretrained video models to scientific simulation is not only a problem of translating video channels into physical variables. It also requires learning a latent representation that is simultaneously accurate enough for physical reconstruction and suitable for downstream generative modeling. The results of this thesis show that these two requirements are related but distinct, and that considering their interaction is essential when transferring latent video generation models to physical dynamics.

Appendix A

Reproducibility Details

This appendix provides supplementary implementation and training details for the experiments reported in Chapters 5 and 6. The model architecture, adaptation methodology, loss formulations, data preprocessing procedure, and evaluation metrics are described in the main text and are therefore not repeated here.

A.1 VAE Training Configuration

Unless otherwise stated, all VAE experiments use the common configuration summarized in Table A.1. Individual experimental arms modify only the settings listed in Section A.2.

Table A.1: Shared VAE fine-tuning configuration used by the baseline.

Setting	Value
Base model	LTX-Video 2B (v0.9.6, dev)
Trainable modules	Entire encoder and decoder
Loss	$\mathcal{L}_{\text{RMSE}} + 50 \mathcal{L}_{H^1}$
Optimizer	AdamW, weight decay 0
Peak learning rate	5×10^{-5}
Warm-up	200 linear steps from $0.01 \times$ peak LR
Scheduler	Cosine decay to 10^{-6}
Epochs	20
Training simulations	4000 per epoch
Effective batch size	32
Gradient clipping	Global norm 5.0
Precision	BF16 mixed precision
Memory optimization	Gradient checkpointing
Checkpoint selection	Lowest validation VRMSE
Random seed	42

A.2 Controlled Experimental Arms

Table A.2 summarizes the controlled VAE experiments. Each arm changes one component of the baseline configuration while the remaining settings are kept fixed. The corresponding adaptation strategies and loss definitions are described in Chapter 4.

Table A.2: Setting changed by each experimental arm relative to the full fine-tuning baseline.

Group	Arm	Change relative to baseline
Adaptation	Unfinetuned	No fine-tuning after four-channel expansion
	Colour adapter	Encoder and decoder frozen; input/output 1×1 adapters trained
	Decoder only	Encoder frozen, including <code>conv_in</code>
	Decoder + <code>conv_in</code>	Encoder frozen except <code>conv_in</code>
Loss	RMSE only	$w_{H^1} = 0$
	RMSE + H^1 + SSIM	$w_{\text{SSIM}} = 0.15$
Learning rate	LR 10^{-5}	Peak learning rate 10^{-5}
Resolution	Trained at 128^2	128×128 training data and statistics
Latent regularization	KL	$\lambda_{\text{KL}} \in \{10^{-7}, 10^{-5}\}$
	SDS	$\lambda_{\text{SDS}} \in \{10^{-4}, 10^{-2}, 1\}$

Two additional runs were started but excluded from the quantitative comparison because training became unstable shortly after the warm-up phase: a VAE trained with a peak learning rate of 10^{-4} and a KL-regularized configuration with $\lambda_{\text{KL}} = 10^{-6}$. These runs are reported only for completeness and are not included in the result tables.

A.3 DiT Training and Inference Configuration

The same DiT training and inference protocol is used for each evaluated VAE. The fixed configuration is summarized in Table A.3.

Table A.3: DiT fine-tuning and inference configuration.

Setting	Value
Trainable modules	Full LTX-Video 2B transformer and γ embedding
Objective	Flow matching
Conditioning	First latent frame and scalar γ
γ embedding	64 Fourier features $\rightarrow$ MLP (hidden size 256) $\rightarrow$ 4 tokens of dimension 4096
Embedding dropout	0.1
Optimizer	AdamW
Learning rate	10^{-5}
Scheduler	Cosine decay to 10^{-8}
Training steps	8000
Effective batch size	32
Gradient clipping	Global norm 1.0
Precision	BF16 mixed precision
Memory optimization	Gradient checkpointing
Inference steps	20
Classifier-free guidance	None (guidance scale 1.0)
Latent sequence	14 latent frames; first frame conditioned, remaining 13 generated
Decoded sequence	105 frames, trimmed to 101 physical frames
Inference seed	42 + simulation index

A.4 Hardware and Runtime

All experiments were conducted on the `sng_pvc` partition using compute nodes equipped with four Intel Data Center GPU Max 1550 cards, exposed as eight accelerator devices. Representative wall-clock times are listed in Table A.4.

Table A.4: Representative wall-clock times for the main training and evaluation jobs on the `sng_pvc` partition.

Job	Time
VAE, full fine-tuning, 256^2 , 20 epochs	15.7–16.7 h
VAE, decoder only, 256^2	8.6 h
VAE, full fine-tuning, 128^2	7.1 h
DiT, 8000 steps	19.5–20.1 h
VAE + DiT evaluation, 500 simulations	1.5 h

Appendix B

Supplementary Dataset and Evaluation Details

B.1 Distribution of the Conditioning Parameter

The experiments use ten discrete values of the heat-capacity ratio γ . The independent test set contains 50 simulations for each value, giving 500 test trajectories in total.

Table B.1: Values of the conditioning parameter γ represented in the test set. Each value contains 50 independent test simulations.

1.13	1.22	1.30	1.33	1.365	1.40	1.404	1.453	1.597	1.76
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The training file contains the same set of γ values, with 450 simulations associated with each value before the training–validation split.

B.2 Normalization Statistics

The normalization procedure is defined in Chapter 5. For reproducibility, Table B.2 reports the training-set statistics used for the primary 256×256 experiments.

Table B.2: Training-set normalization statistics for the four physical fields at 256×256 resolution.

Field	Mean μ_c	Standard deviation σ_c
Density ρ	0.905	0.662
Momentum m_x	0.004	0.534
Momentum m_y	−0.002	0.527
Pressure p	0.903	0.805

The corresponding statistics at 128×128 resolution differ by less than 0.2%.

B.3 Uncertainty Estimation

The quantitative test metrics are evaluated independently for each of the 500 test simulations. The reported standard error of the mean (SEM) is

$$\text{SEM} = \frac{s}{\sqrt{N}}, \quad N = 500, \quad (\text{B.1})$$

where s denotes the sample standard deviation of the per-simulation metric. The SEM therefore quantifies variation across the sampled test trajectories; it does not account for variation caused by retraining the model with different random seeds.

B.4 Run-to-Run Variation

A limited set of repeated experiments was used to estimate the scale of training variability. Repeating the baseline configuration with three random seeds at 128×128 resolution produced a spread of approximately 1.1% in validation VRMSE.

This observation provides only an approximate indication of run-to-run variability and should not be interpreted as a systematic multi-seed uncertainty analysis. Differences on the order of 1% between experimental arms should therefore be interpreted cautiously rather than as robust improvements.

Appendix C

Complete Numerical Results

This appendix reports the complete numerical results underlying the comparisons presented in Chapter 6.

Table C.1 reports VAE reconstruction performance on the independent test set. Table C.2 provides the corresponding validation results and enables comparison of validation and test behaviour. Table C.3 reports the downstream DiT rollout performance for the evaluated VAE configurations.

Table C.1: VAE reconstruction VRMSE on the held-out test set (500 simulations, 256×256). Δ is the change of the channel-mean VRMSE relative to the baseline. Lower is better. The model trained at 128^2 is applied directly to the 256×256 test data (the VAE is fully convolutional).

Arm	Pooled	Ch. mean $\pm$ SEM	Δ	ρ	m_x	m_y	p
<i>Adaptation</i>							
Unfinetuned (inflated only)	0.4774	0.5569 ± 0.0117	+595.9%	0.1269	0.1116	0.1302	1.8589
Colour adapter	0.4081	0.4643 ± 0.0102	+480.2%	0.1050	0.0863	0.0924	1.5736
Decoder only	0.1061	0.1441 ± 0.0023	+80.0%	0.0974	0.0870	0.0938	0.2980
Decoder + conv_in	0.0677	0.0929 ± 0.0015	+16.1%	0.0843	0.0772	0.0803	0.1299
Full FT (baseline)	0.0590	0.0800 ± 0.0014	–	0.0727	0.0707	0.0704	0.1064
<i>Loss</i>							
RMSE only	0.0603	0.0814 ± 0.0014	+1.7%	0.0767	0.0718	0.0724	0.1048
RMSE + H1 + SSIM (0.15)	0.0656	0.0889 ± 0.0015	+11.1%	0.0808	0.0787	0.0786	0.1174
<i>Learning rate</i>							
LR 1×10^{-5}	0.0683	0.0938 ± 0.0015	+17.2%	0.0805	0.0773	0.0778	0.1395
<i>Resolution</i>							
Trained at 128^2	0.0882	0.1200 ± 0.0019	+49.9%	0.1202	0.1009	0.1016	0.1572
<i>Latent reg.</i>							
KL 10^{-7}	0.0593	0.0804 ± 0.0014	+0.5%	0.0728	0.0707	0.0705	0.1077
KL 10^{-5}	0.0650	0.0877 ± 0.0014	+9.6%	0.0807	0.0792	0.0784	0.1126
SDS 10^{-4}	0.0585	0.0793 ± 0.0014	–0.9%	0.0723	0.0703	0.0700	0.1046
SDS 10^{-2}	0.0595	0.0806 ± 0.0014	+0.8%	0.0734	0.0714	0.0709	0.1069
SDS 1	0.0766	0.1040 ± 0.0015	+29.9%	0.0963	0.0941	0.0940	0.1316

Table C.2: VAE reconstruction VRMSE on the validation split (500 simulations held out from the training file) at the checkpoint with the best validation VRMSE. Δ_{val} and Δ_{test} compare each arm with the baseline on the validation and test sets. The model trained at 128^2 is validated on 128^2 data.

Arm	Ep.	Pooled	Ch. mean	Δ_{val}	Δ_{test}	ρ	m_x	m_y	p
<i>Adaptation</i>									
Colour adapter	20	0.4042	0.4691	+493.0%	+480.2%	0.1031	0.0846	0.0921	1.5967
Decoder only	20	0.1013	0.1403	+77.4%	+80.0%	0.0956	0.0867	0.0938	0.2853
Decoder + conv_in	20	0.0658	0.0918	+16.0%	+16.1%	0.0821	0.0766	0.0804	0.1280
Full FT (baseline)	20	0.0575	0.0791	—	—	0.0709	0.0700	0.0705	0.1051
<i>Loss</i>									
RMSE only	20	0.0587	0.0805	+1.8%	+1.7%	0.0748	0.0713	0.0726	0.1036
RMSE + H1 + SSIM (0.15)	20	0.0639	0.0879	+11.1%	+11.1%	0.0787	0.0780	0.0787	0.1161
<i>Learning rate</i>									
LR 1×10^{-5}	20	0.0666	0.0928	+17.3%	+17.2%	0.0785	0.0765	0.0780	0.1382
<i>Resolution</i>									
Trained at 128^2	20	0.0871	0.1181	+49.3%	+49.9%	0.1096	0.1083	0.1085	0.1461
<i>Latent reg.</i>									
KL 10^{-7}	20	0.0578	0.0795	+0.5%	+0.5%	0.0710	0.0700	0.0707	0.1064
KL 10^{-5}	20	0.0633	0.0868	+9.8%	+9.6%	0.0788	0.0787	0.0785	0.1113
SDS 10^{-4}	20	0.0570	0.0784	−0.9%	−0.9%	0.0705	0.0696	0.0702	0.1034
SDS 10^{-2}	20	0.0580	0.0797	+0.8%	+0.8%	0.0717	0.0708	0.0710	0.1056
SDS 1	20	0.0746	0.1029	+30.0%	+29.9%	0.0943	0.0933	0.0939	0.1299

Table C.3: Channel-mean VRMSE of the VAE reconstruction alone and of the full pipeline (encode frame 0, DiT rollout, decode) on the test set (500 simulations), for a DiT trained for 8k steps on each VAE’s latents. Δ is relative to the baseline VAE; the per-channel columns are the VAE + DiT errors; *Latent* is the VRMSE of the rolled-out latent against the encoder’s latent of the ground truth.

VAE	VAE only	VAE + DiT $\pm$ SEM	Δ	ρ	m_x	m_y	p	Latent
Full FT (baseline)	0.0802	0.4984 $\pm$ 0.0060	—	0.514	0.506	0.516	0.459	0.558
Unfinetuned (inflated only)	0.5579	0.9501 $\pm$ 0.0128	+90.6%	0.637	0.646	0.657	1.859	0.583
KL 10^{-7}	0.0807	0.4891 $\pm$ 0.0059	−1.9%	0.507	0.499	0.506	0.445	0.558
KL 10^{-5}	0.0879	0.4733 $\pm$ 0.0053	−5.0%	0.493	0.480	0.488	0.433	0.273
SDS 10^{-4}	0.0795	0.4974 $\pm$ 0.0060	−0.2%	0.512	0.504	0.514	0.459	0.556
SDS 10^{-2}	0.0808	0.4583 $\pm$ 0.0054	−8.1%	0.483	0.465	0.474	0.411	0.290
SDS 1	0.1044	0.5678 $\pm$ 0.0051	+13.9%	0.572	0.578	0.585	0.536	0.065

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Disclaimer

I hereby declare that this thesis is entirely the result of my own work except where otherwise indicated.
I have only used the resources given in the list of references.

Garching, September 27, 2026

(Signature)